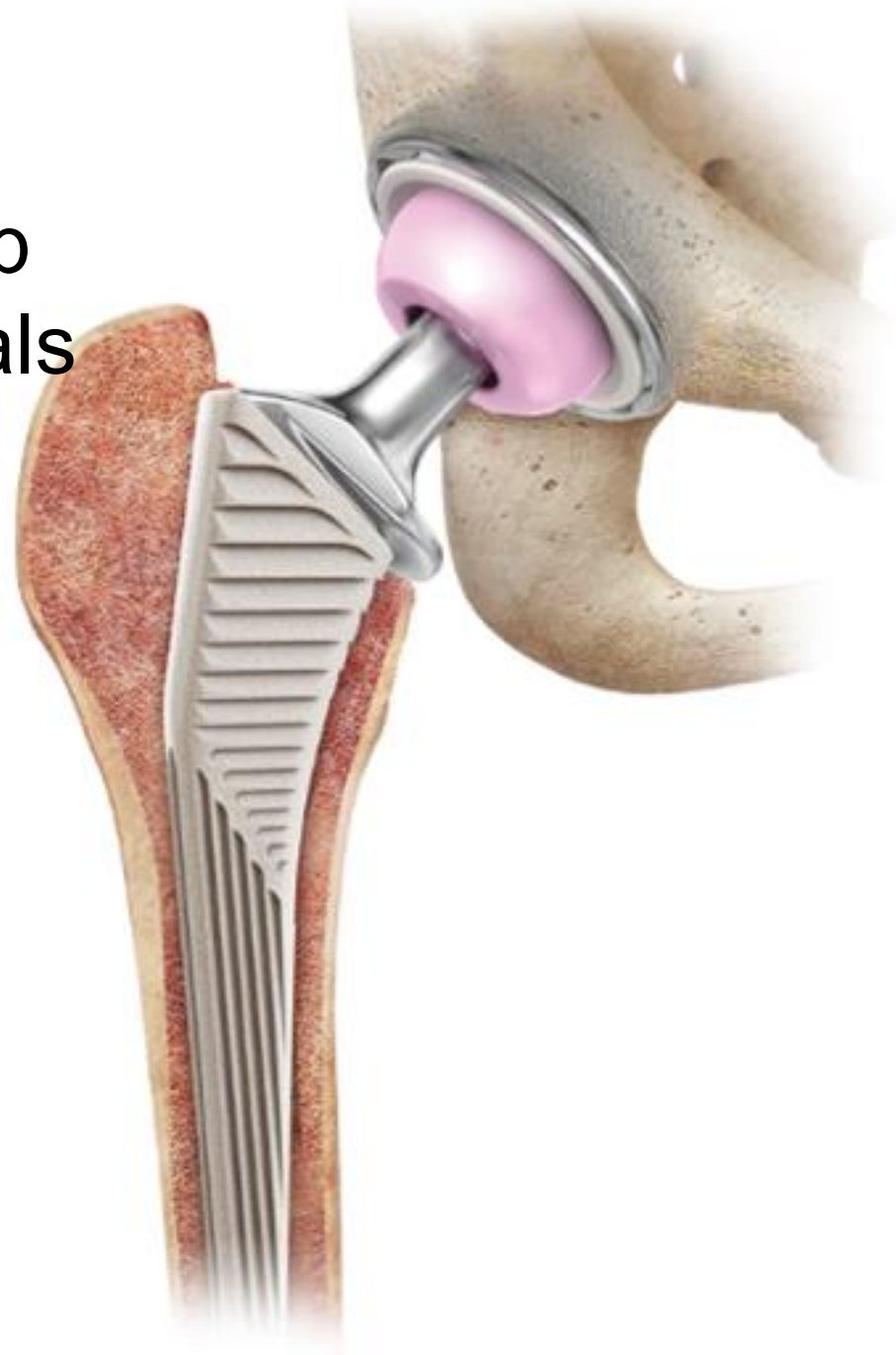
Applications and Advances of Zirconia-Toughened Alumina Ceramics in Total Hip Arthroplasty: From High-Strength Materials to Predictable 20-Year Reliability

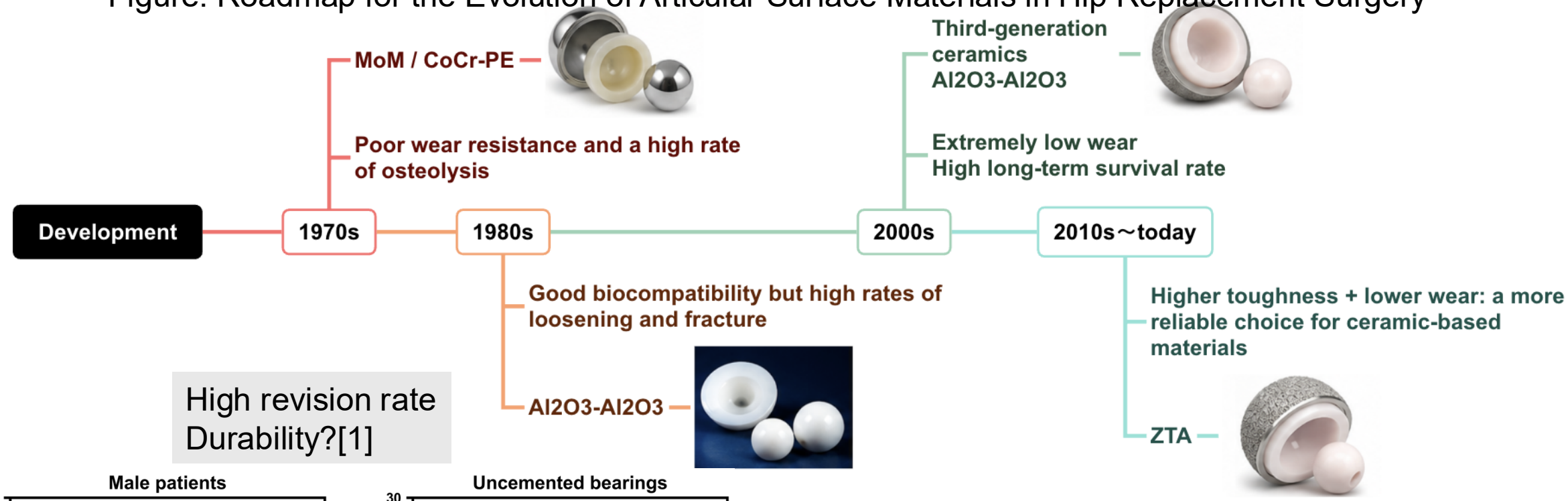


Alumina 5			
Student name	**QM student number**	**NPU student number**	**Permission*Yes or No**
Xuanming Che			yes
Haotai Xu			yes
Xiangjun Zheng			yes
Yongqi Xu			yes

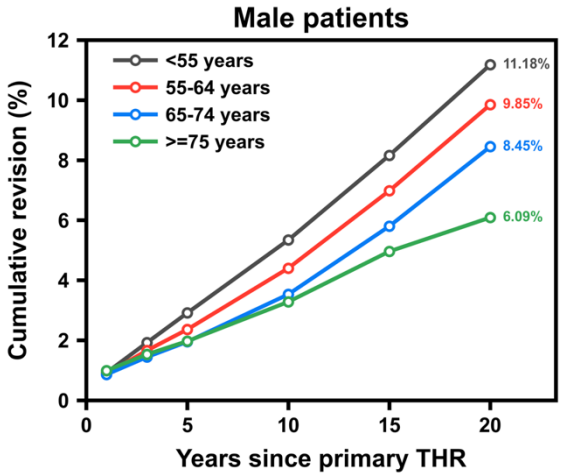
Xuanming Che	
Introduction	

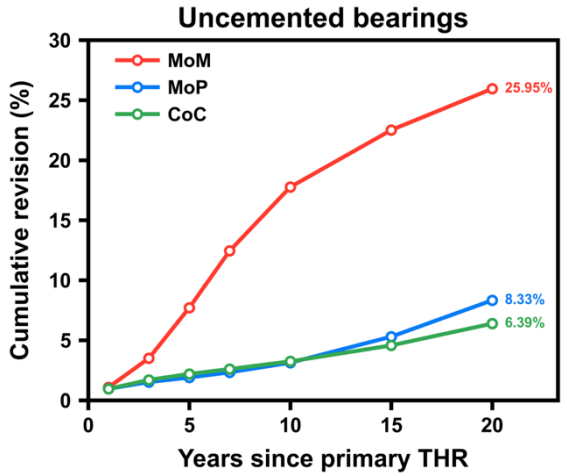
Strength Isn't the End Goal: Can We Predict Reliability Over 20 Years?

Figure. Roadmap for the Evolution of Articular Surface Materials in Hip Replacement Surgery



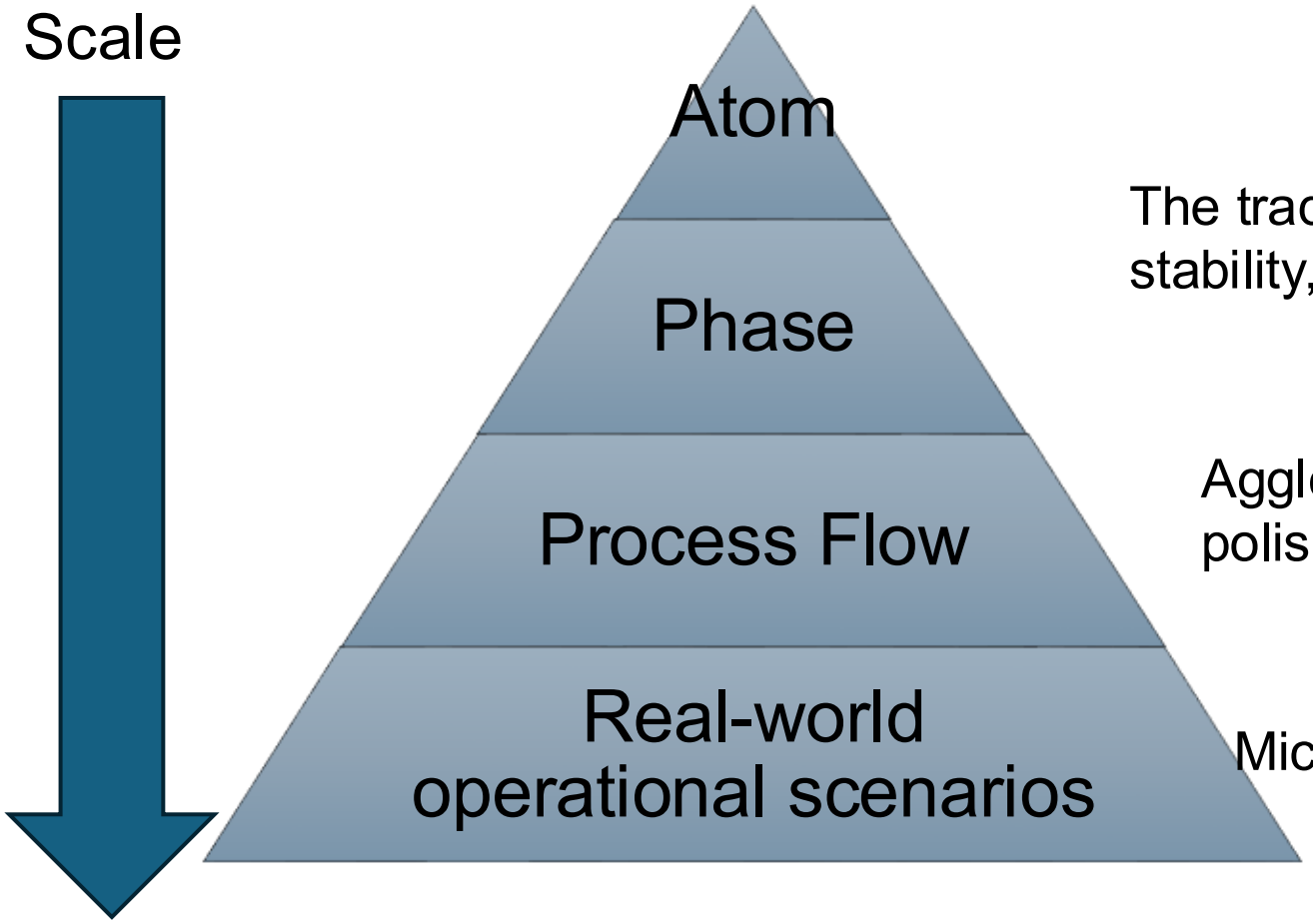
High revision rate
Durability?[1]





With a hardness of up to 1,600 HV[2], the question today is not “Is it strong?” but “Can we predict its reliability for more than twenty years before implantation?”

From inception to maturity, four types of constraints collectively define ZTA's long-term reliability

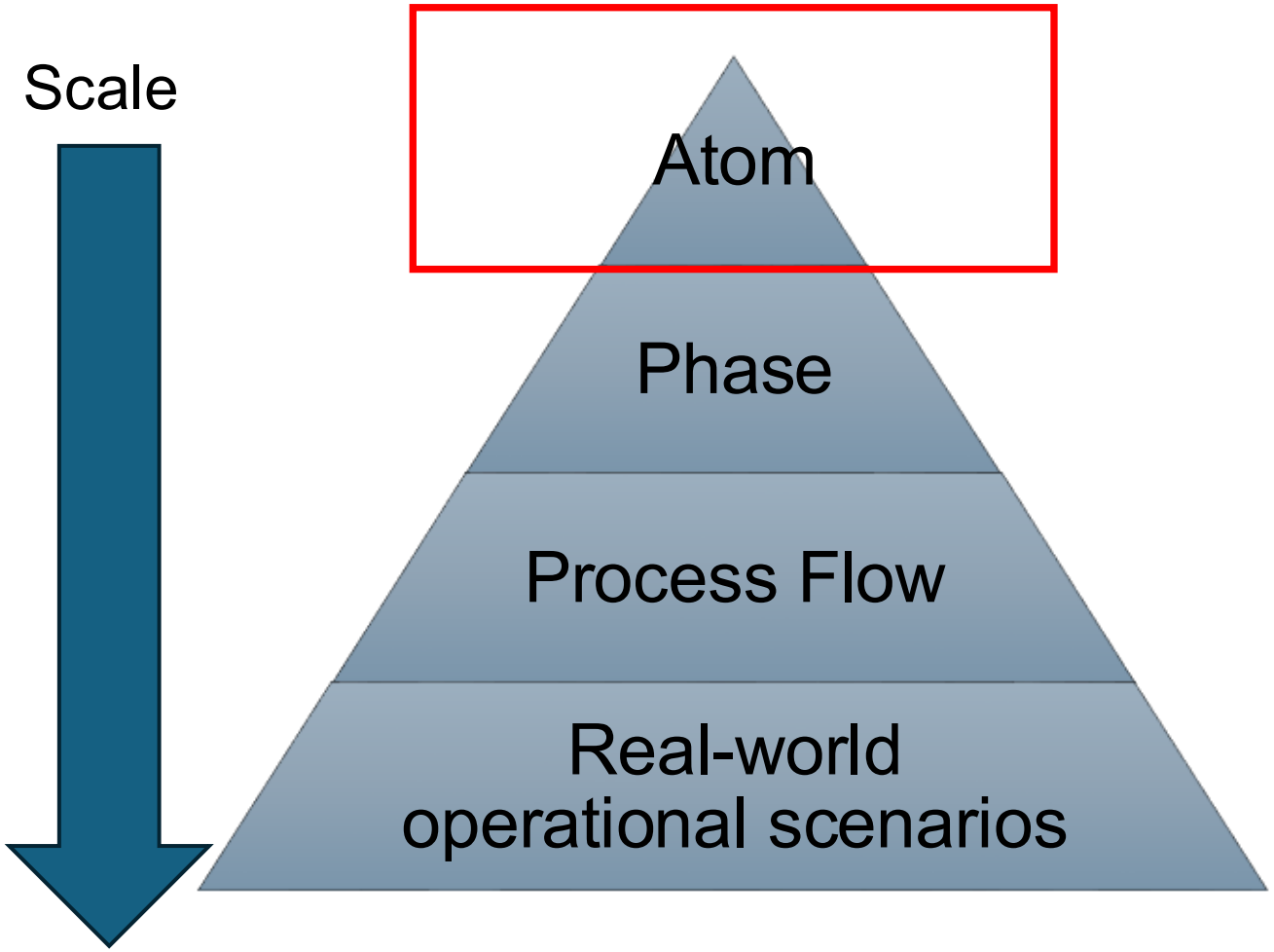


The trade-off between toughening and phase stability, and the low-strength tail

Agglomeration of powders, sintering/HIP, polishing damage

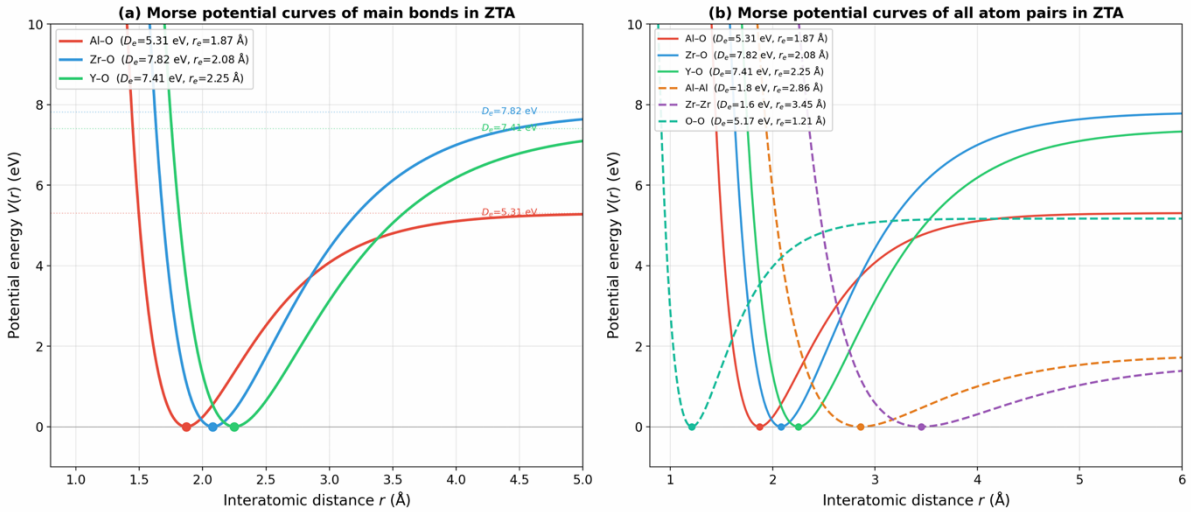
Micro-separation, edge contact, wear, fracture

Hao Tai Xu	
Key Issues	



Hao Tai Xu	
Key Issues	

Atomic bonding: strong bonds, high stiffness, low ductility



Morse Potential: Ceramic vs Metal

The narrower the range of bond lengths for metal-O bonds, the stronger the bond between atoms (the higher the hardness), but the more difficult it is to deform them (the greater the brittleness).

	Work hardening	Solid solution hardening	Precipitation hardening	Hall-Petch
Alloy	✓	✓	✓	✓
Pure Alumina	✗	✗	✗	✓
ZTA	✗	✓	✓	✓

ZTA can conduct more strengthening mechanisms than pure alumina, with high hardness

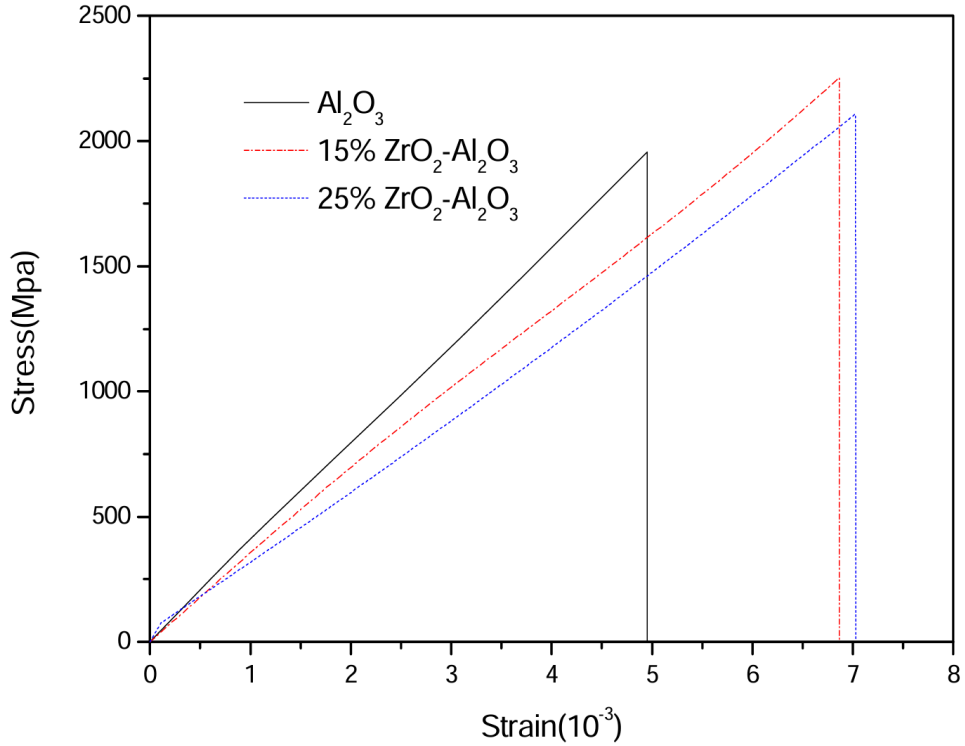
Pauling Ionic Properties

Al–O : 56.7%

Zr–O : 67.1%

Y–O : 70.8%

High ionic character + covalent orientation
= steep potential well
= high stiffness, low ductility



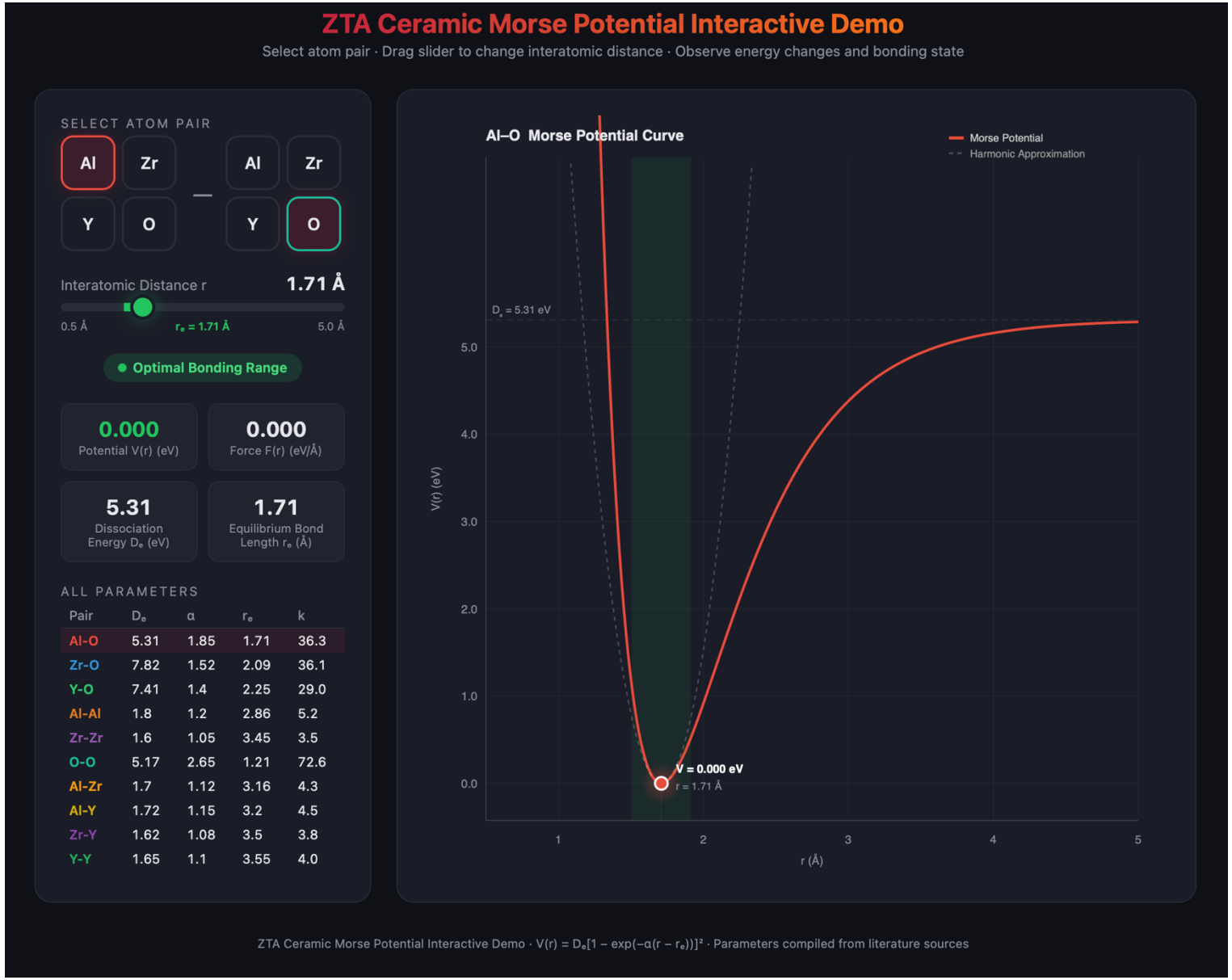
Compression loading curve. Zirconia contributes to strength, although reduce modulus. [1]

Interactive: Adjust the interatomic distance to observe the trade-off between strong bonds and low plasticity

Alumina5

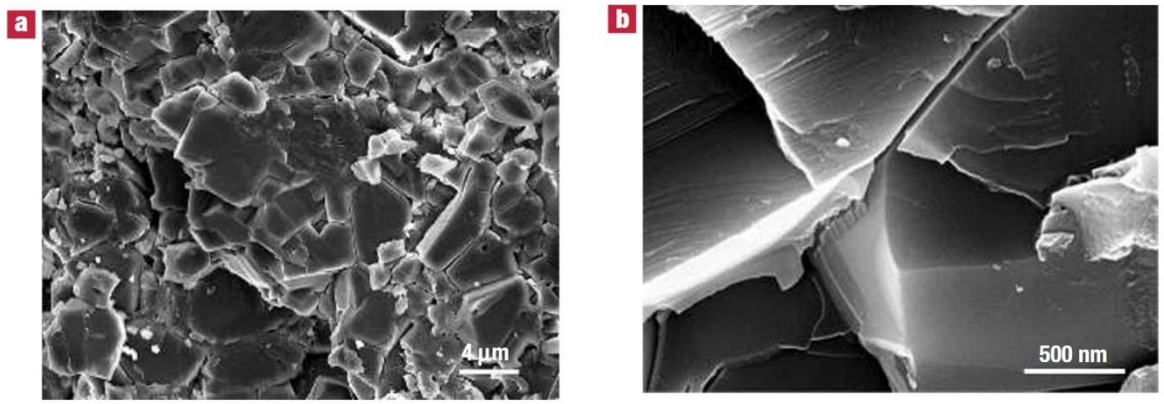
Hao Tai Xu	
Key Issues	

Interactive Demonstration

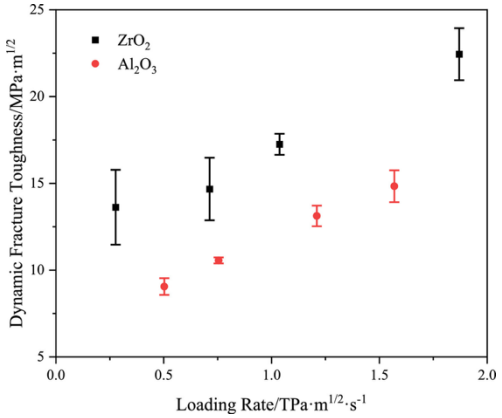


Hao Tai Xu	
Key Issues	

From Alumina to ZTA: Why Hardness Alone Isn't Enough



Fracture morphology of AD995 alumina fragments under a 7.8 GPa impact load. a, Scanning electron microscope (SEM) micrograph of a large fragment. b, Cleavage steps on the fracture surface at high magnification. [2]



Higher dynamic fracture toughness for Zirconia, a suitable toughening phase.[3]

Alumina: Hard but Brittle

- Dynamic frac. toughness 8~15 MPa√m
- Excellent wear resistance
- No plastic deformation

Zirconia: Toughened phase

- Dynamic frac. toughness ~ 12-25 MPa√m
- Particle based Hall-Petch

Remaining Concern

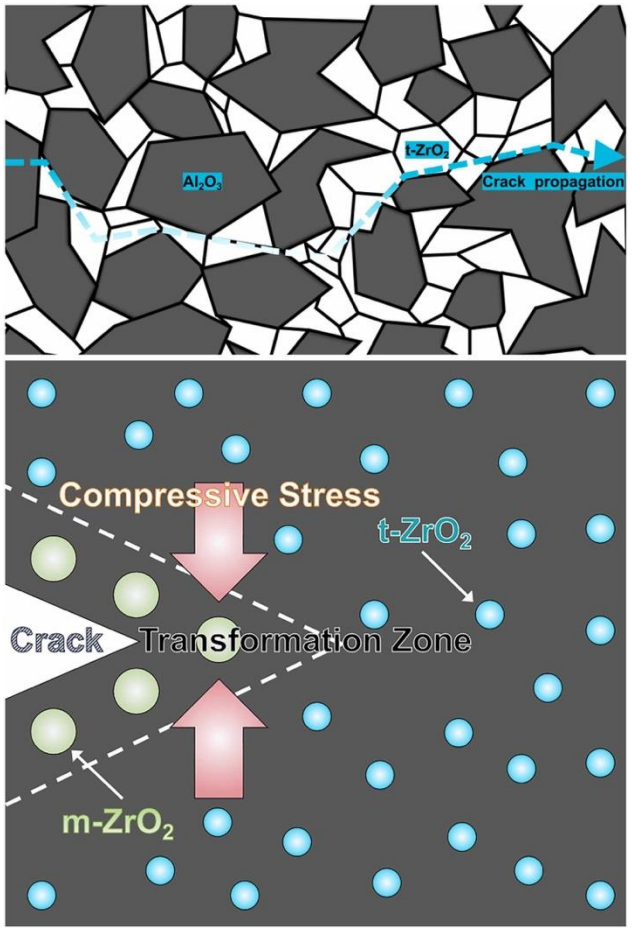
Strong bonding = no plastic flow

- Reliability depends on flaw & phase control

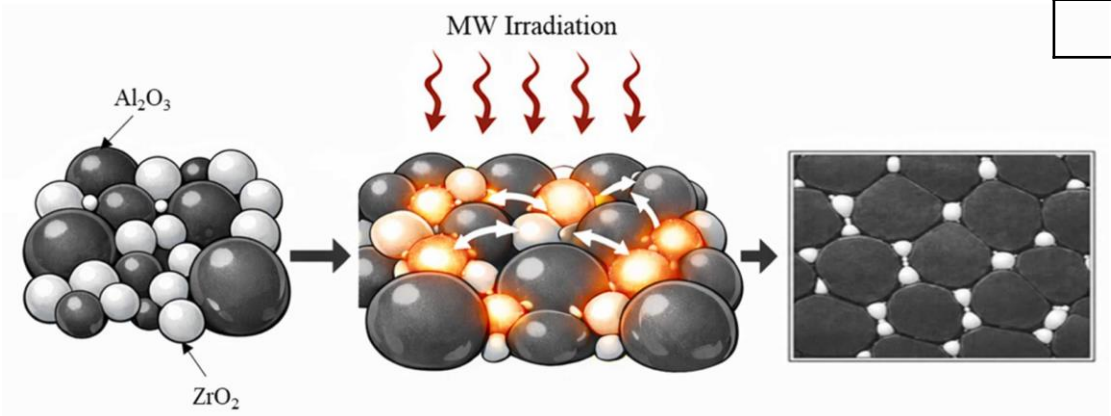
Hardness alone is insufficient; transformation toughening bridges the gap, but phase stability remains the critical challenge.

Hao Tai Xu	
Key Issues	

How Far Can Transformation Toughening Be Safely Used?

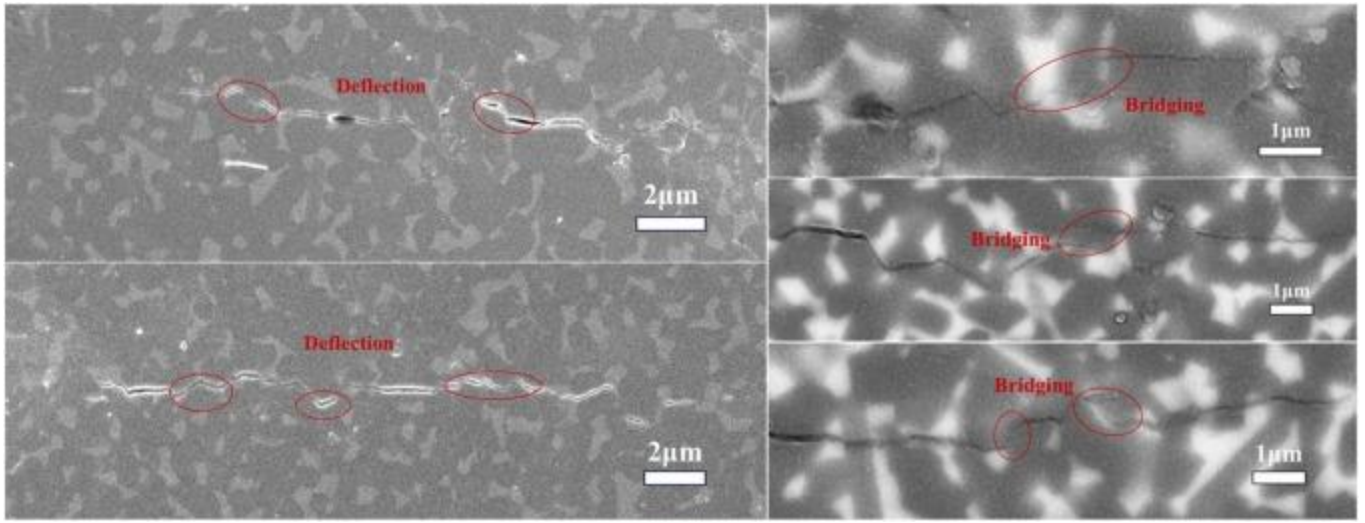


Schematic diagram of stress-induced t→m-ZrO2 phase transformation toughening mechanism.[4]



Schematic diagram illustrating the formation mechanism of the relatively uniform distribution structure in ZTA ceramics during microwave sintering. [5]

The safe toughening window is bounded by insufficient toughening on one side and spontaneous transformation / LTD on the other.



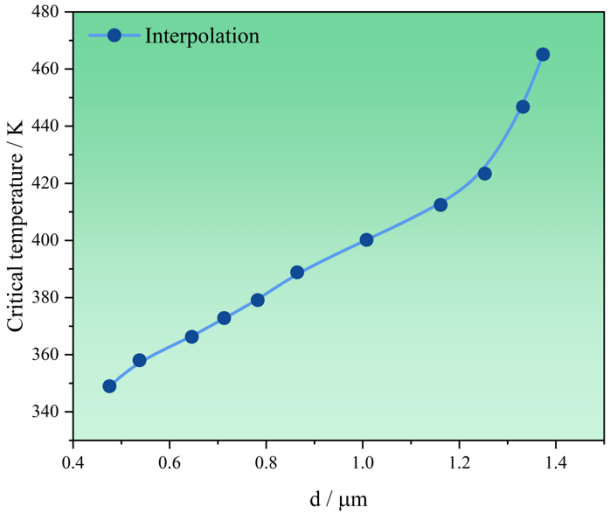
ZrO2 Briging at cracks[5]

Phase transformation limited in near-crack region, ZrO2 resist propagation (transformation and bridging), avoid whole-body failure

Hao Tai Xu

Emerging Trends

Emerging Trend: Tuning the Useful ZrO_2 Metastability Window



$$M_s = \frac{0.34 \times 10^6 d^2 - 5.49 \times 10^3 d - 4.97 d^{0.5} - 8.69 \times 10^{-3}}{7.34 \times 10^3 d^2 + 0.565 d}$$

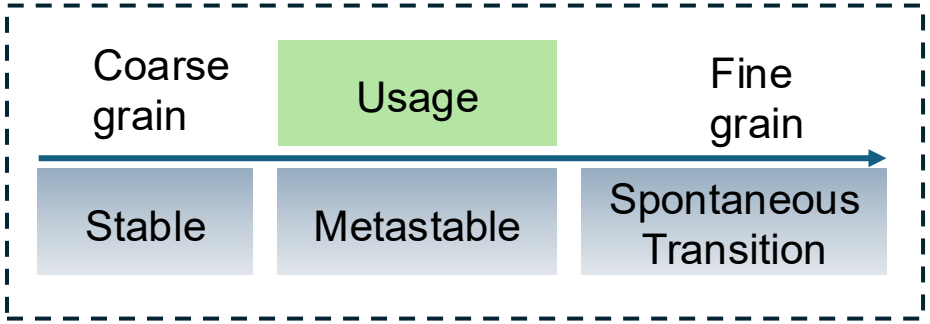
Zhu (1995) [6]Provided a interpolation formula for the critical phase transformation of ZrO_2 (add Y_2O_3 2 at.%)

Easier transition with smaller grain, lower strength with larger grain (Hall-Petch)

Pure alumina joint — Past

ZTA joint — Now

Future



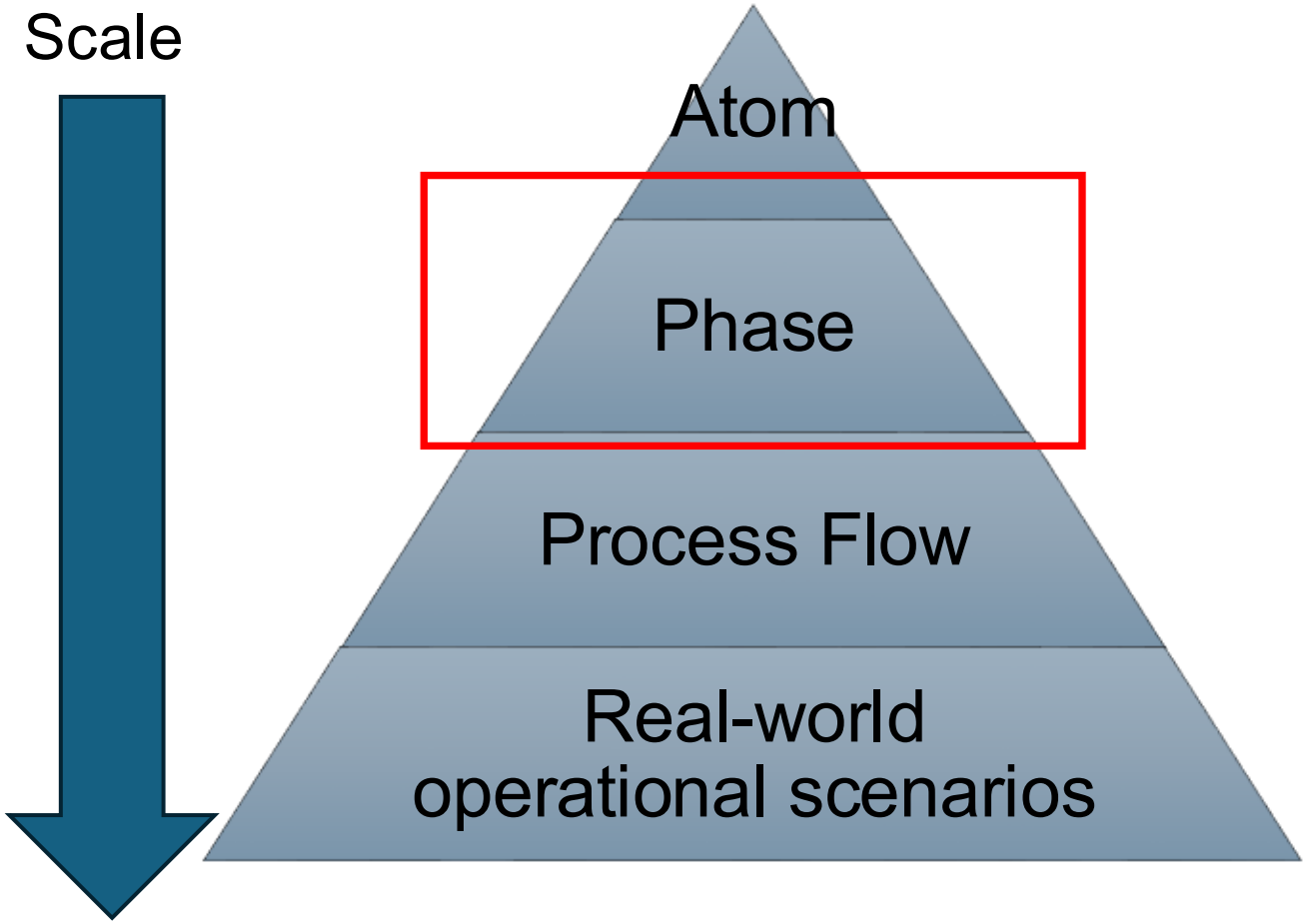
ZrO_2 Metastability Window in ZTA

Personalized component:
Adapt special body fluid condition, e.g. Diabetes
Life cycle in-situ monitor: Predict degradation, wearing, etc.
Sustainable manufacturing: e.g. Low sintering temperature

Key Parameters: ZrO2 grain size (0.2-0.6 μm optimal) | Y2O3 (1.5-3 mol%) | Al2O3 matrix constraint | Cooling rate | Service temperature

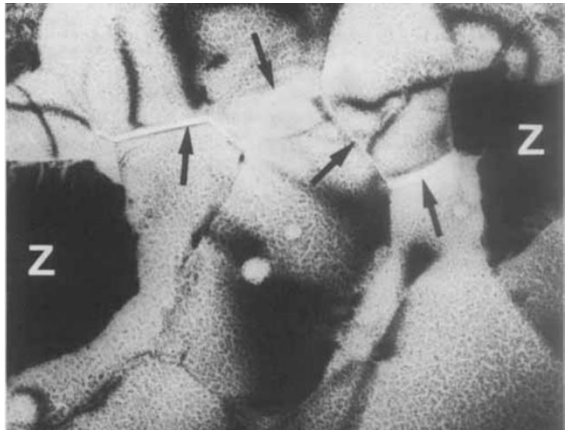
From empirical selection to physics-based, computationally guided tuning of the ZrO2 metastability window.

Xiangjun Zheng	
Key Issues	



Xiangjun Zheng	
Key Issues	

Is it a useful metastable state or an early phase transition?

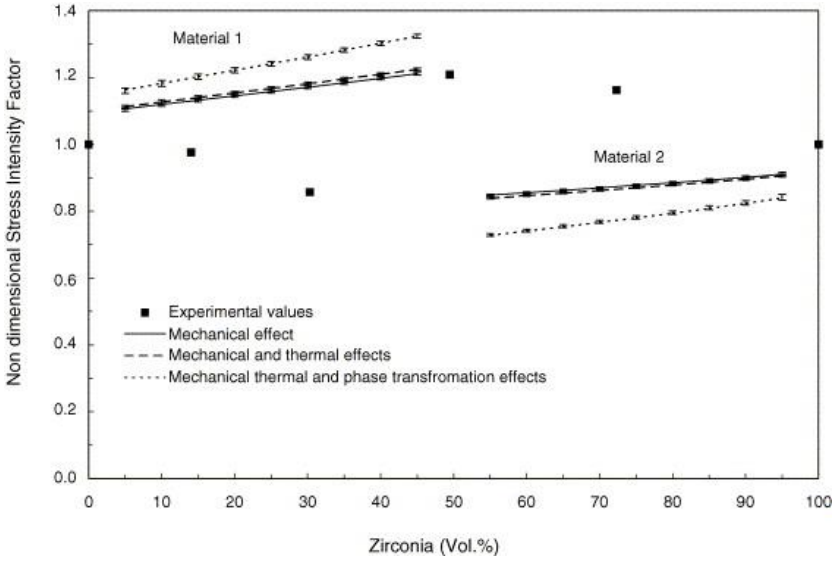


TEM Morphology of Phase-Transition-Induced Microcracks: Illustrating the Microstructural Basis of Toughening[1]

01 Desired Metastability

The t→m phase transition is precisely triggered at the crack tip, releasing phase transition energy.

The volumetric expansion resulting from the phase transformation creates a compressive stress field, effectively closing cracks and toughening the material.

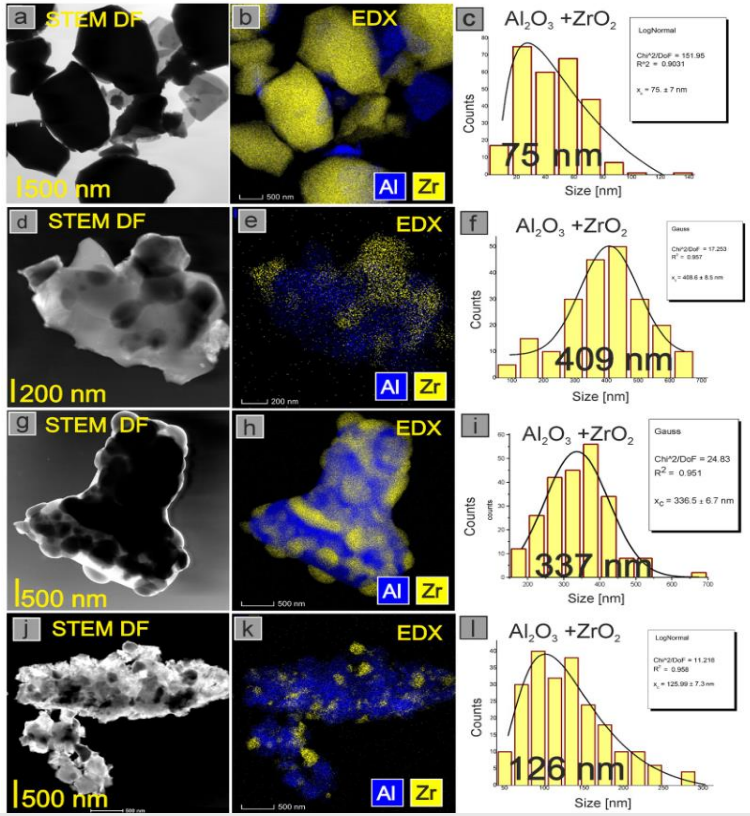


Curve showing the effect of ZrO_2 content on toughness[2].

02 Risk of premature phase change

Spontaneous phase transformation prior to service leads to surface roughening and microcrack propagation.

This triggers low-temperature degradation (LTD), significantly reducing the material's mechanical properties.[2]



Schematic illustration of the nano-isolation effect, offering a new approach to regulating phase stability. [3]

03 Core Regulation of Metastability: The Nano-Isolation Effect

Is it possible to develop a multi-factor coupled model to predict phase stability in complex in vivo environments?[3]

Xiangjun Zheng

Key Issues

m/t/c-ZrO_2: Structure Determines Stability

01 Thermodynamic Scale

At room temperature, the monoclinic (m) phase is the most stable, while the tetragonal/cubic (t/c) phases are metastable. Doping with Y_2O_3 to introduce oxygen vacancies can effectively reduce the free energy of the t phase, allowing it to be retained at room temperature. However, caution must be exercised regarding the spinodal decomposition region, as the immense chemical driving force can lead to the spontaneous breakdown of the metastable state, triggering phase separation.[4]

Y_2O_3 doping

↓

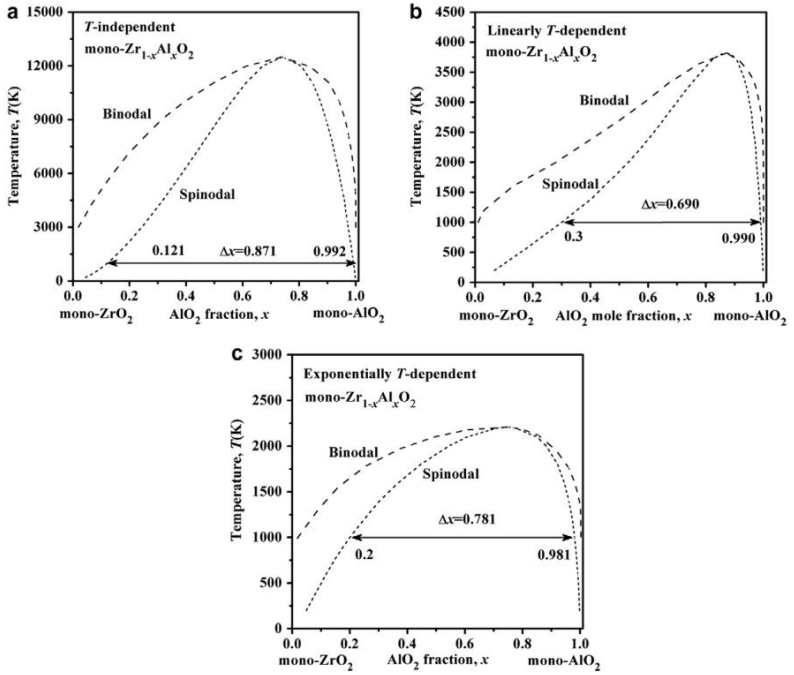
Formation of oxygen vacancies

↓

Reduce the free energy of t-ZrO_2

↓

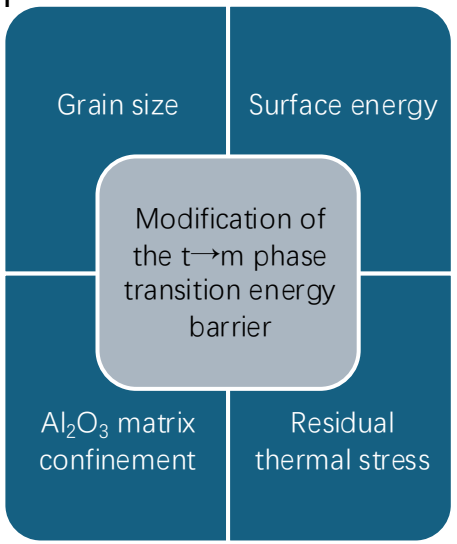
Retention of metastable t-ZrO_2 at room temperature



02 Kinetic Barrier

Metastability is essentially the result of a balance between thermodynamics and kinetics. Grain size, surface energy, and the chemical and physical constraints of the Al_2O_3 matrix collectively form the phase transformation energy barrier. Residual thermal stress significantly influences the critical conditions for the t→m phase transformation by altering the crack driving force, making it a key parameter that must be quantified in engineering applications.

Inclusion ZrO_2 (%)	K_I/K_0		
	$a = r_m/100$	$a = r_m/50$	$a = r_m/25$
5	1.111	1.107	1.100
10	1.123	1.120	1.114
15	1.135	1.133	1.127
20	1.148	1.145	1.140
25	1.161	1.158	1.154
30	1.174	1.171	1.167
35	1.187	1.184	1.180
40	1.200	1.198	1.194
45	1.214	1.212	1.208



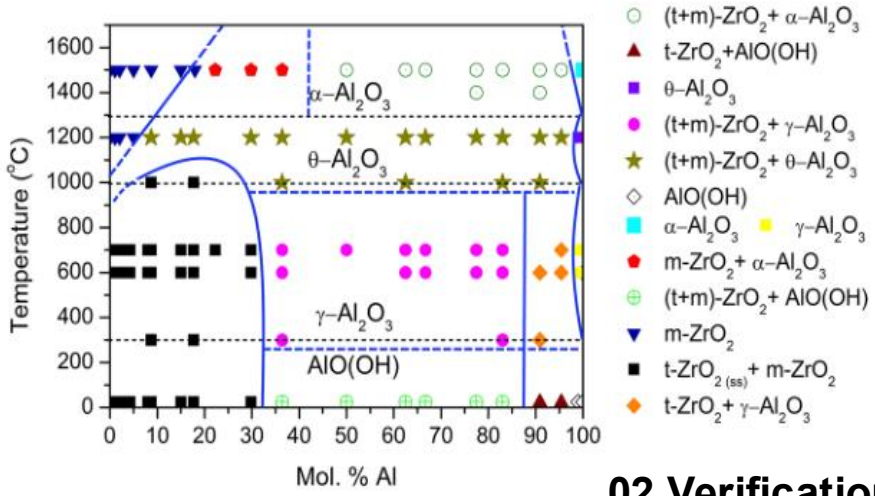
Cesari (2006) [2]quantified the regulatory effect of residual thermal stress on the stress intensity factor (K). The confinement effect of the matrix not only alters the phase transformation temperature but also directly determines the material's crack resistance and structural integrity under loading conditions

Xiangjun Zheng

Emerging Trends

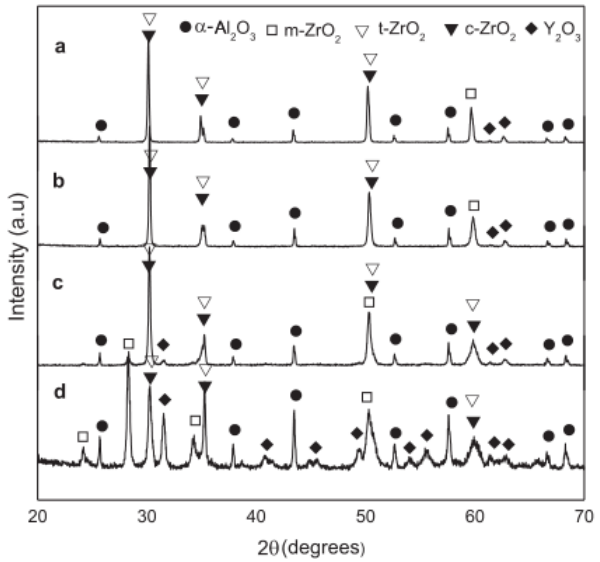
Doping, oxygen vacancies, and Al_2O_3 confinement collectively determine stability

01 Nanoscale effects



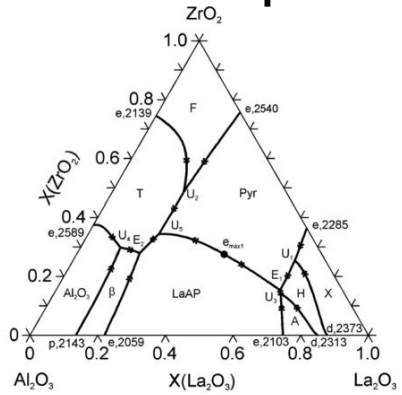
The high surface energy of nanoparticles combined with the Al_2O_3 barrier effect reshapes the phase equilibrium and broadens the stability range of t-ZrO_2.[3]

02 Verification of Phase Evolution



XRD analysis confirms that 4–6 wt% Y_2O_3 represents the optimal range, with the m phase virtually disappearing and the t phase reaching its optimal proportion.[5]

03 Multi-component co-doping design



The ternary phase diagram guides the co-doping of multiple rare earth elements, allowing for precise control of oxygen vacancy concentration to enhance phase stability and toughening effects.[6]

04 Macro-level Performance Validation

Contents of Y_2O_3 (wt%)	Phase present	Lattice parameter				Phase content (%)	Hardness HV (MPa)	Hardness Knoop (MPa)	Fracture toughness (MPa $m^{1/2}$)
		a (nm)	*b* (nm)	*c* (nm)	*β* (°)				
2	α-Al_2O_3	0.4990	0.4990	0.8183		52.02	11.30	10.83	2.53
	m-ZrO_2	0.5397	0.4071	0.5877	98.96	31.24			
	t-ZrO_2	0.5935	0.5935	0.4930		8.58			
	c-ZrO_2	0.5153	0.5153	0.5153		2.51			
4	α-Al_2O_3	0.5128	0.5128	0.7527		50.43	13.61	12.87	3.25
	m-ZrO_2	0.5402	0.4075	0.5886	99.01	32.59			
	t-ZrO_2	0.5086	0.5086	0.5110		10.00			
	c-ZrO_2	0.5034	0.5034	0.5034		3.72			
6	α-Al_2O_3	0.4131	0.4131	0.7365		50.50	14.73	16.02	4.82
	m-ZrO_2	0.5149	0.5250	0.5990	99.00	32.55			
	t-ZrO_2	0.5247	0.5247	0.5229		10.22			
	c-ZrO_2	0.5497	0.5497	0.5497		3.34			
8	α-Al_2O_3	0.4990	0.4990	0.8183		51.20	14.07	15.58	4.41
	m-ZrO_2	0.5399	0.5298	0.5753	98.52	35.84			
	t-ZrO_2	0.5099	0.5099	0.5183		10.03			
	c-ZrO_2	0.5147	0.5147	0.5147		1.50			

A Y_2O_3 content of 4–6 wt% represents the "optimal range" for performance, with the composite exhibiting a hardness of 14.73 GPa and a fracture toughness of 4.82 MPa·$m^{1/2}$, confirming the effectiveness of the design strategy.[5]

Xiangjun Zheng

Emerging Trends

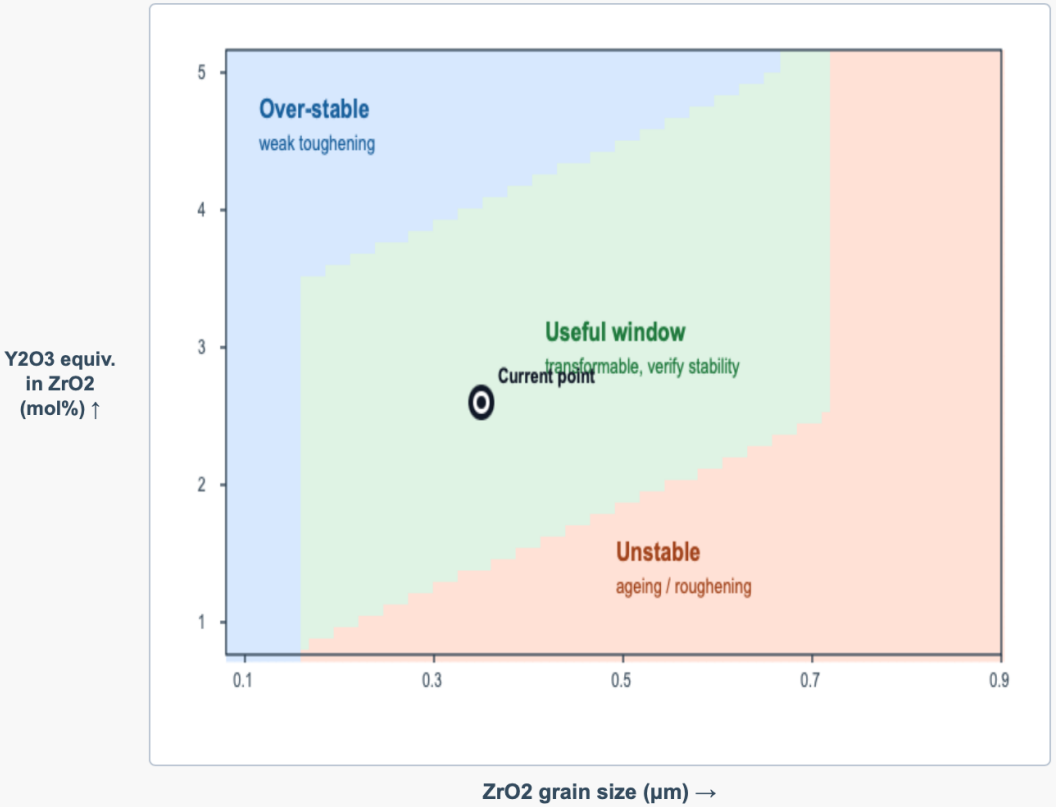
Interactive Demonstration

Doping, oxygen vacancies, and Al_2O_3 confinement collectively determine stability

Phase Stability Window

Emerging Trend / Interactive

Grain size, Y2O3 and Al2O3 constraint tune t-ZrO2 stability



Over-stable
weak toughening

Useful window
verify

Unstable
ageing risk

ZrO2 grain size:

0.35

μm

Y2O3 equivalent:

2.60

mol%

Al2O3 constraint

low | **medium** | high

Useful window

Toughening		90
Ageing		8
Validation		52

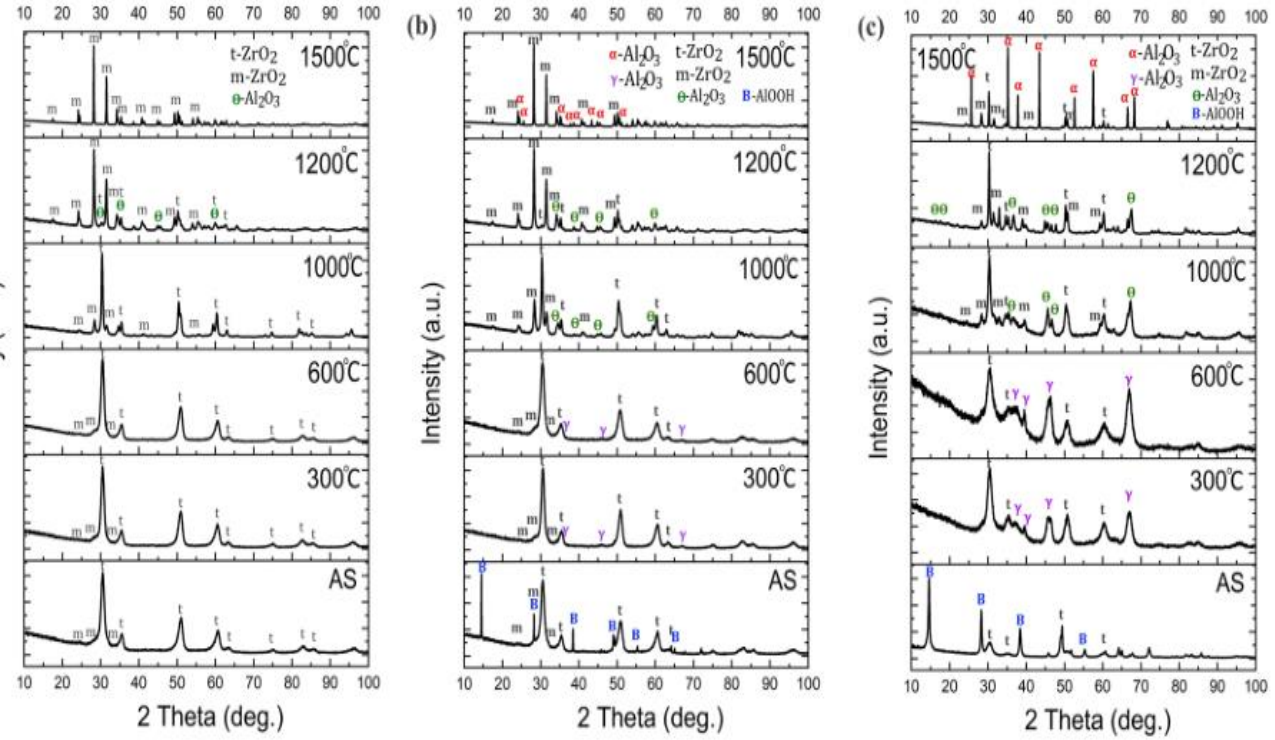
Validation: Verify with Raman mapping and ageing tests.

Interactive: Xiangjun Zheng

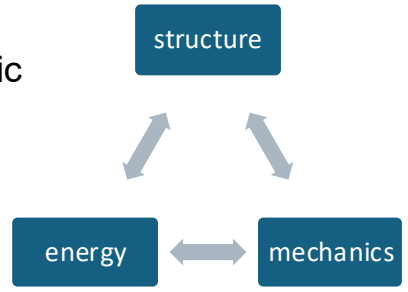
Xiangjun Zheng

Phase stability must be measured; it cannot simply be assumed.

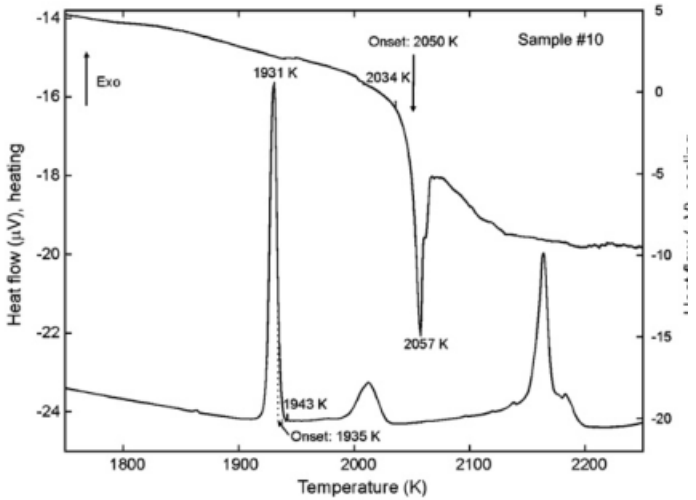
01 Quantitative Characterization by Temperature-Programmed XRD



From macroscopic XRD phase evolution and thermodynamic DTA energy changes to microscopic SEM fracture morphology, we have established a comprehensive validation chain encompassing "structure, energy, and mechanics," ensuring the scientific rigor and accuracy of high-temperature stability evaluations for materials.[3]

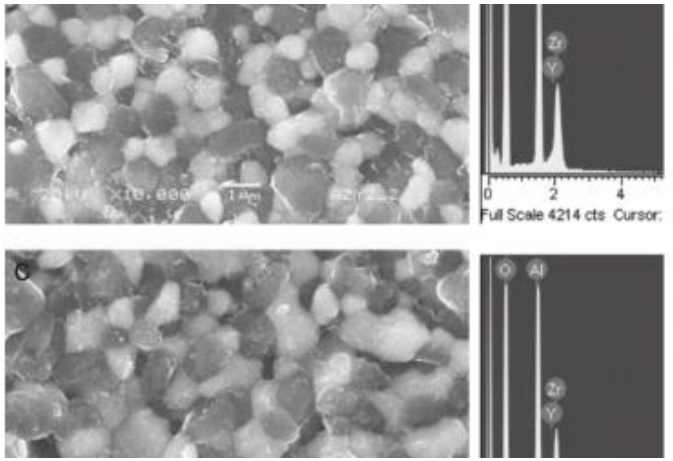


02 DTA Thermal Analysis Validation



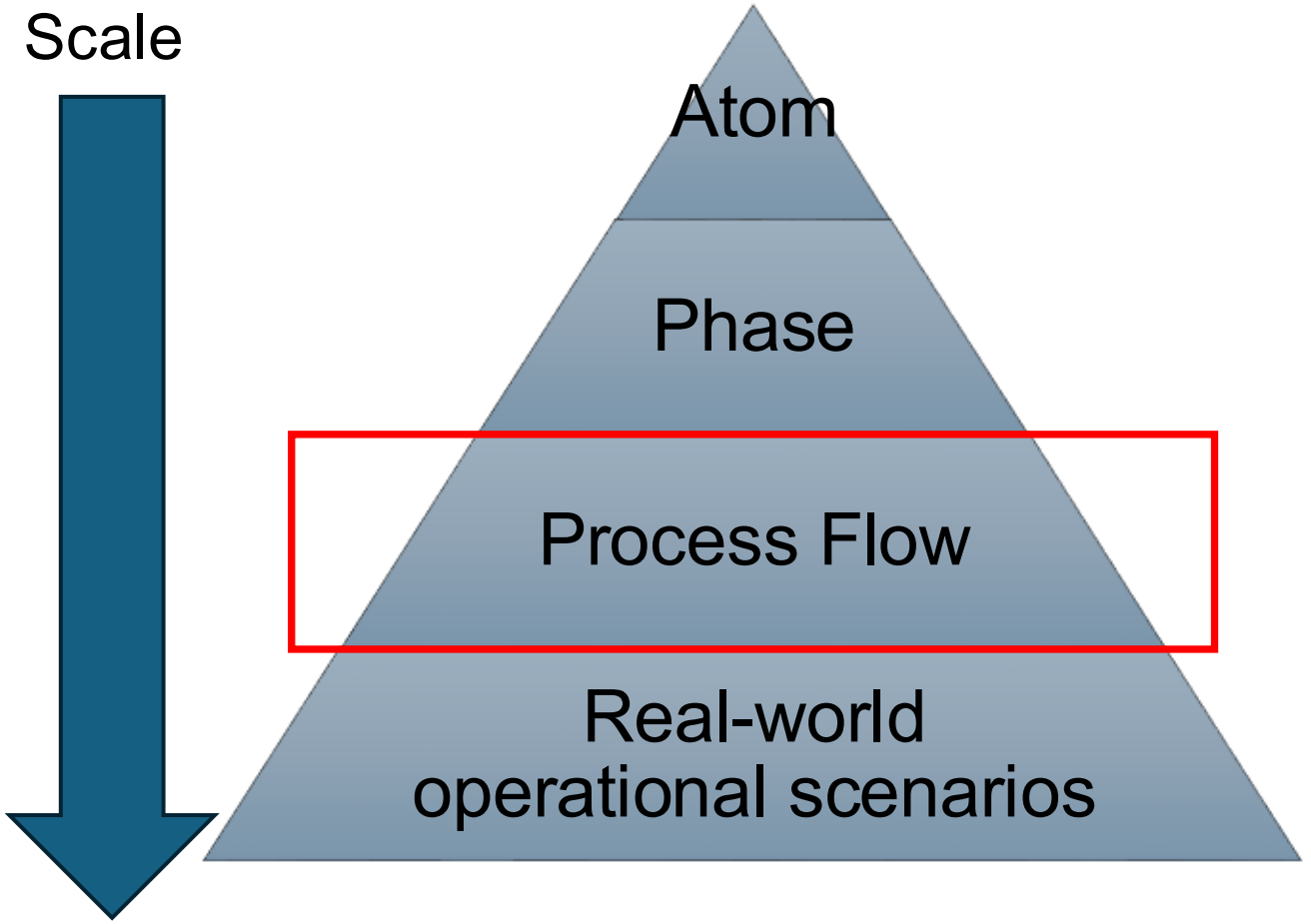
DTA captures the heat absorption and release peaks of phase transitions, accurately determines the phase transition temperatures and enthalpy changes, and verifies the accuracy of the thermodynamic model.[6]

03 SEM microscopic analysis of the fracture surface



SEM images of fracture surfaces clearly reveal toughening phenomena such as crack deflection and grain pull-out, confirming the correlation between microstructure and mechanical properties.[5]

Yongqi Xu	
Key Issues	



Processing – When Does It Become a Reliability & Sustainability Risk?

Yongqi Xu	
Key Issues	

Preparation of mixed powder → **Green body preparation** → **Firing and heat Treatment** → **Polishment**

Powder agglomeration[1]

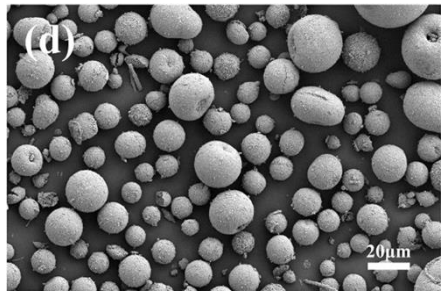
Green density gradient[2]

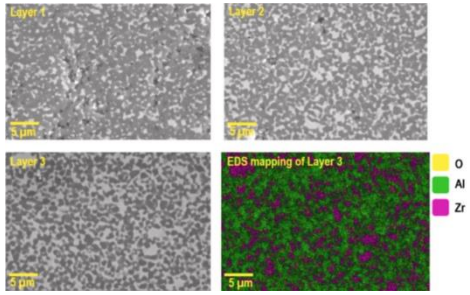
Pores / abnormal grains[3]

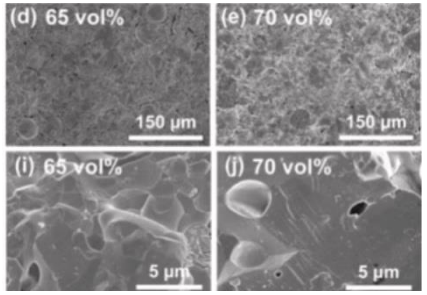
Microcracks[4]

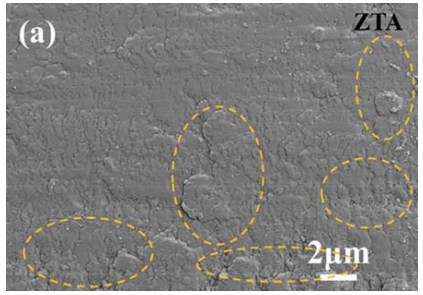
High cost and energy
Sintering and HIP cost too much energy

Failed to manufacture
scale-up consistency and batch variability





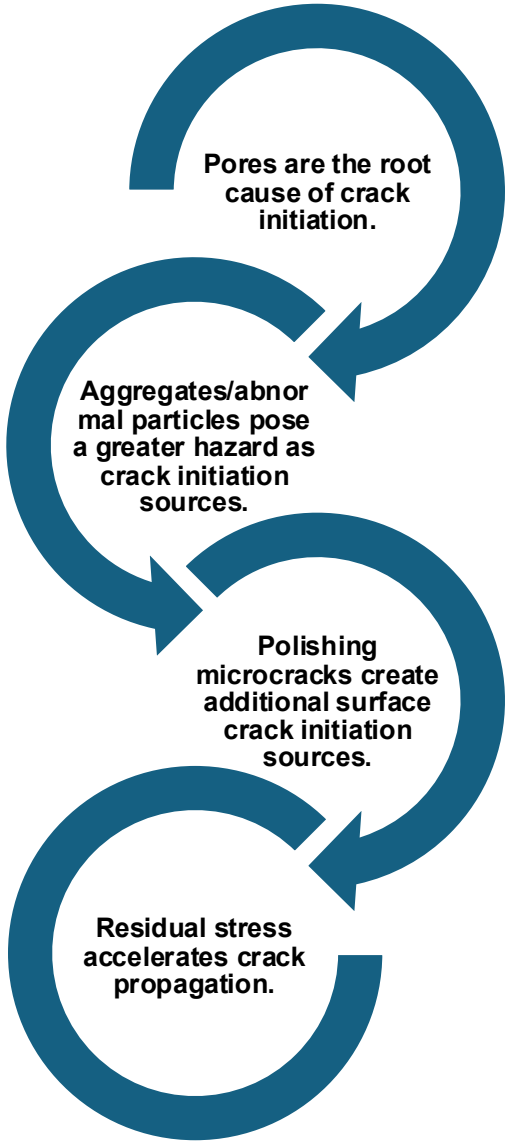


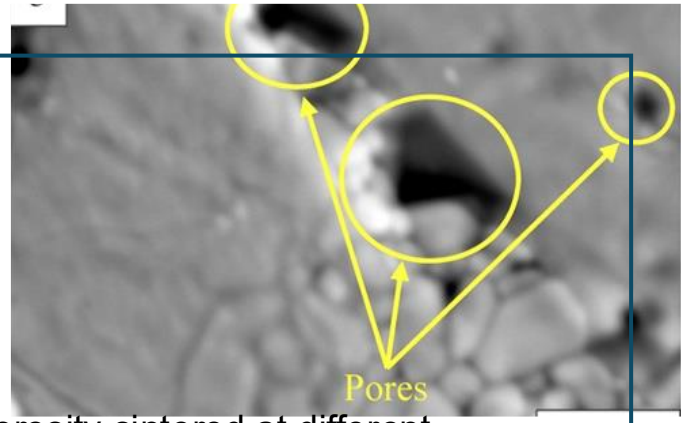


Processing Defects – Pores, Agglomerates, Residual Stress, Polishing Damage

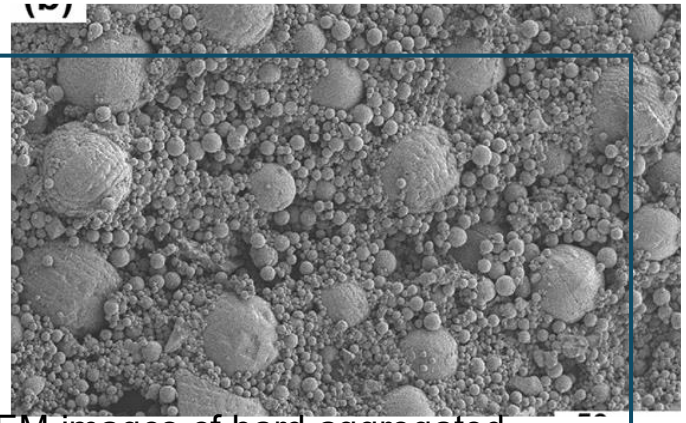
Yongqi Xu

Key Issues

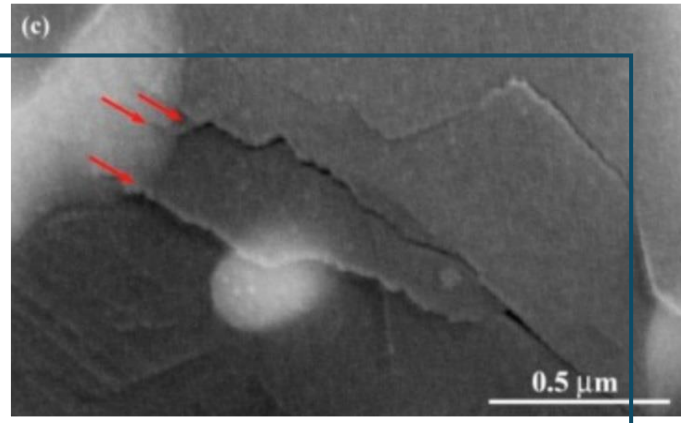




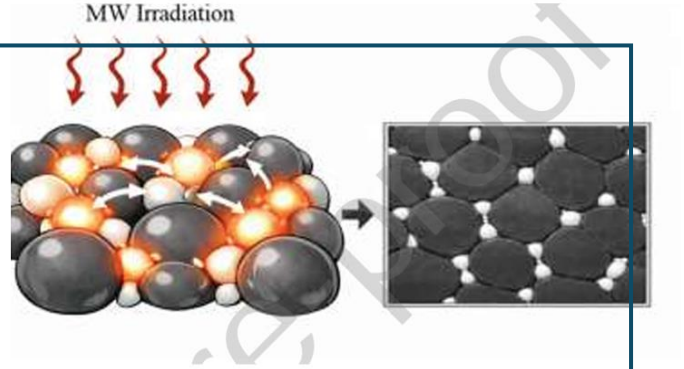
Porosity sintered at different temperatures [5]



SEM images of hard aggregated groups [6]



SEM image of microcracks [7]



The effect of residual stress on the structure after sintering [8]

Emerging Processing Trends – From Wet Dispersion to HIP & Beyond

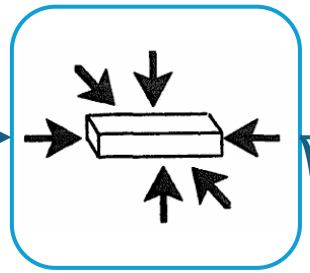
Yongqi Xu

Emerging Trends

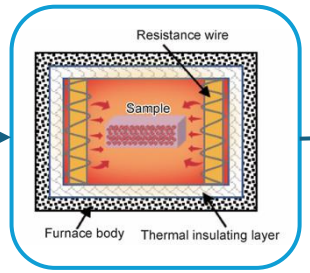
Recent improvements



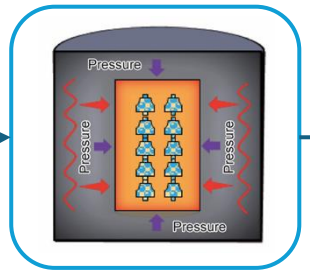
Wet ball milling[12]



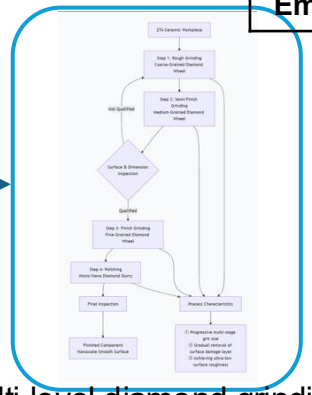
CIP[14]



Pressureless sintering[12]

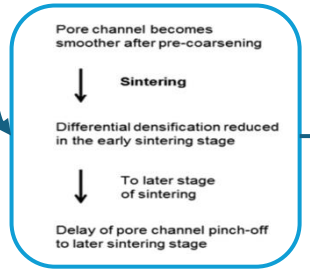


HIP[12]

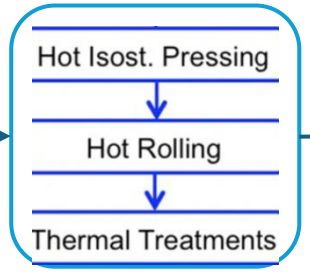


Multi level diamond grinding[13]

Research Stage Route



Two step sintering[10]



HIP cycle optimization[9]

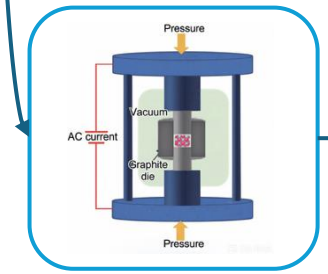
- White light interferometer: surface roughness detection, Ra < 0.02 μm
- SEM: Surface Defect Detection

polishing QA

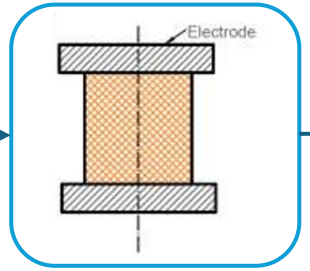
Advantages	Risk/Limitations
Grain control, low defect	Verification required

Mature Biomedical Route

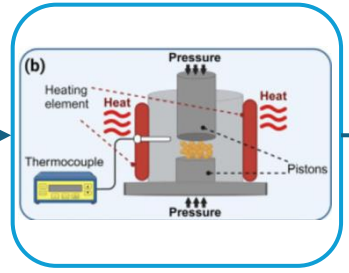
Advantages	Risk/Limitations
Low energy consumption, rapid densification	Difficulty in certification and risk of scaling up



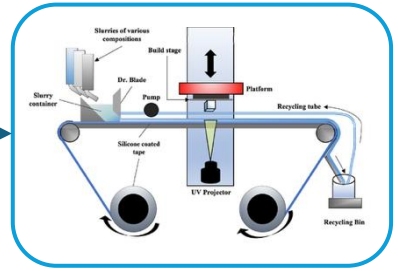
SPS[12]



Flash sintering[15]



cold sintering[9]



additive manufacturing[11]

High cost, long cycle, single dense structure → Short process, low energy consumption, controllable structure, and customizable performance

Interactive Process Window – Density, Grain Size, Transformability, Energy

Yongqi Xu

Emerging Trends

Sintering Temperature Slider

1300°C ← → 1600°C

Current: 1600 °C

Holding Time Slider

0 h ← → 12 h

Current: 5.1 h

Output Indicators

Relative Density Risk: **HIGH (>97% but over-sintered)**

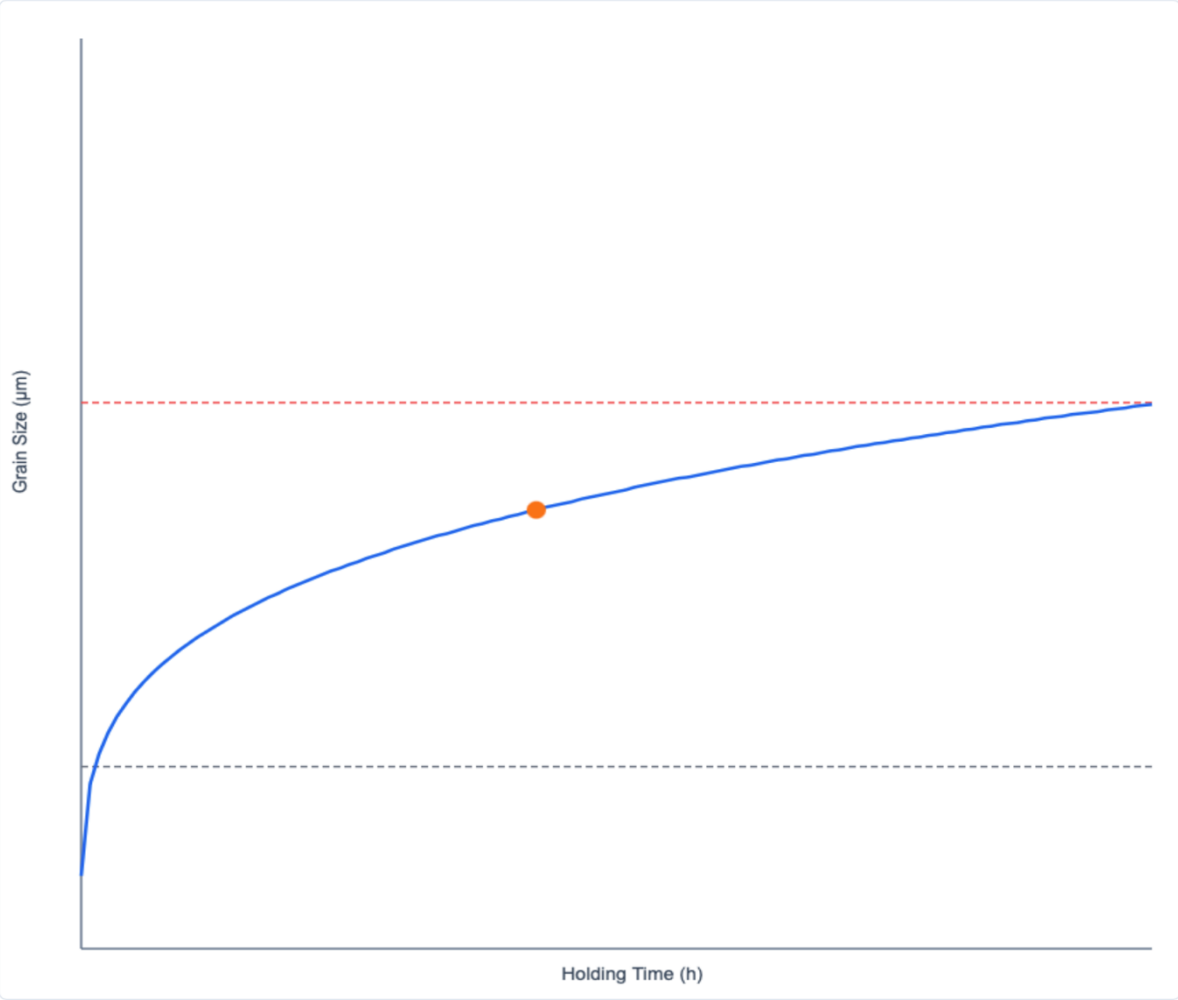
ZrO_2 Grain Size: 0.482 μm (**Optimal transformable range**)

Transformable Risk Zone: **High transformability**

Relative Energy Burden: **HIGH**

ZrO_2 Grain Size vs Holding Time Curve

Valid transformable grain range: 0.2~0.6 μm



Process QA – Capturing Failure Risk & Environmental Cost in Every Measurement

QA measurement indicators

- SEM grain size
- Density/porosity
- XRD phase ratio
- Surface roughness

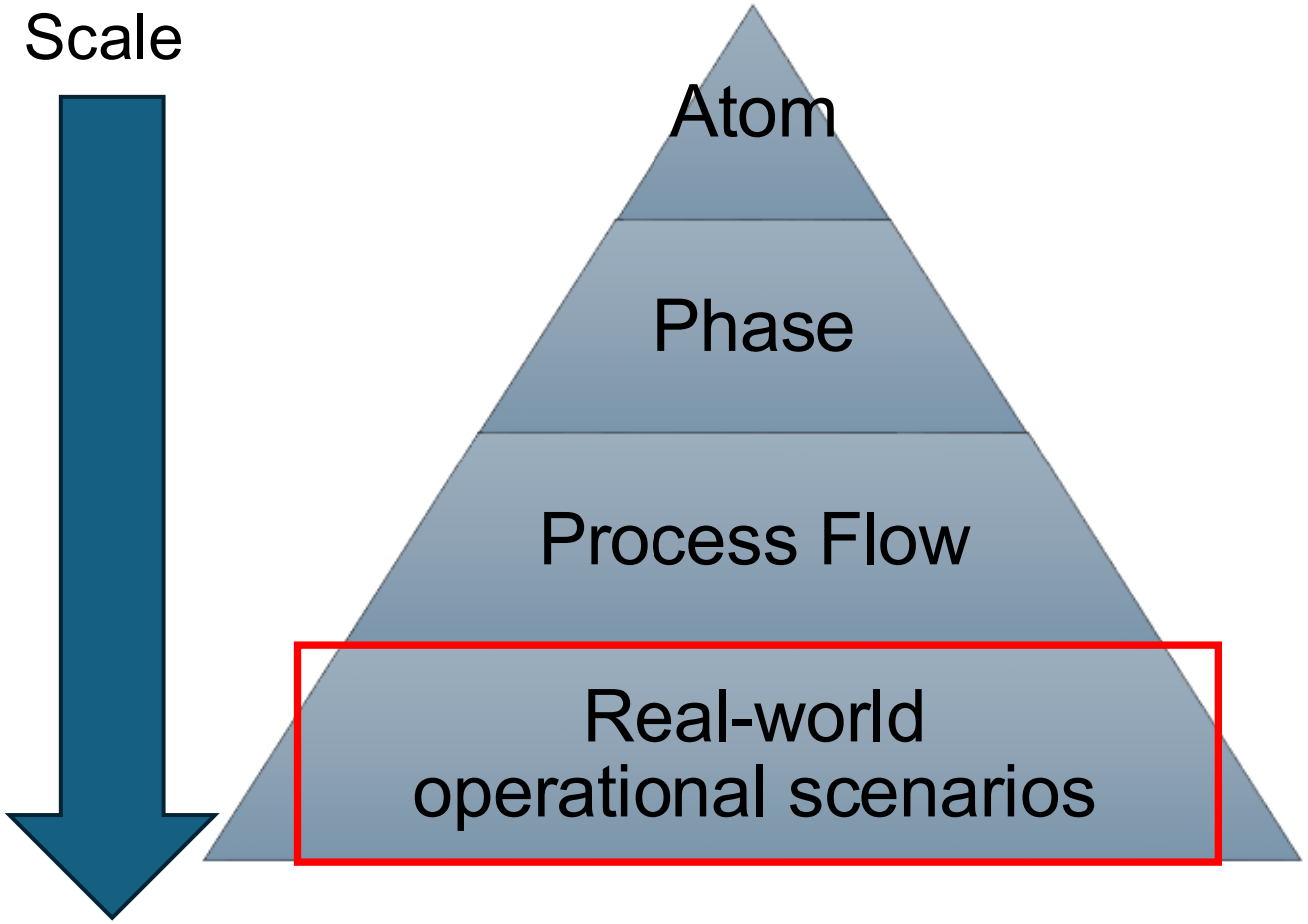
Corresponding failure mechanism

- ZrO_2 exceeds toughening window → loses toughening/spontaneous phase transformation
- Pores → crack initiation
- Early phase transition → LTD risk
- Polishing microcracks → fatigue source

Quantification of Sustainability

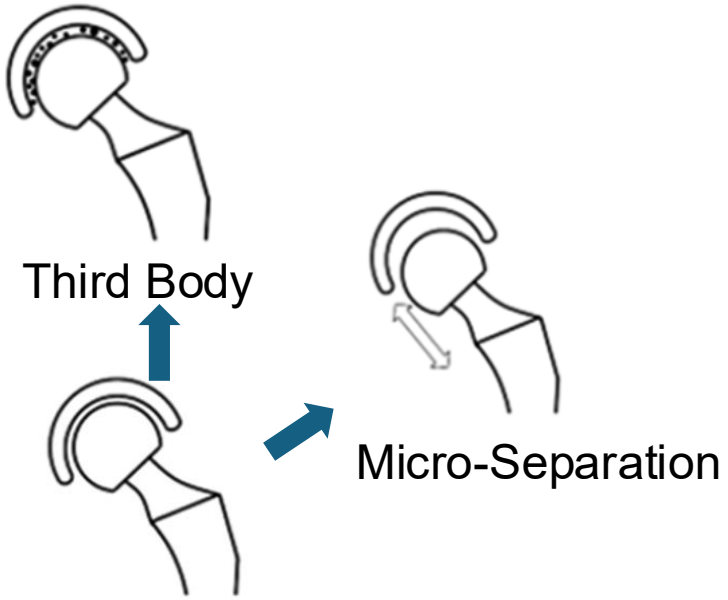
- Energy consumption: 1488.6 kWh/qualified parts[16]
- Y_2O_3 dosage: 1.9 mol%[17]

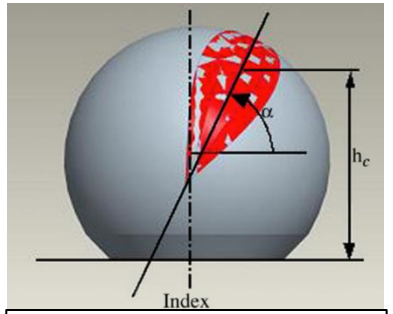
Xuanming Che	
Key Issues	



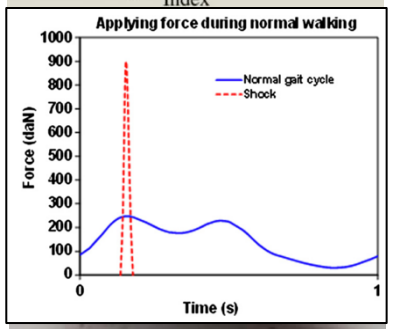
Xuanming Che
Key Issues

In actual service, localized stress is a key factor





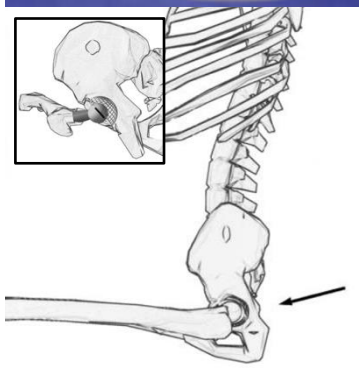
Schematic illustration of striation wear (side view), generated by a procedure proven to produce striation wear consistent with that observed during implant removal. [1]



Loads experienced during walking and impact.[4]



Third-body particles embedded in the inner bearing[3]



Rear edge loading. When standing up from a chair or climbing high steps, high loads are generated on the surface of the head.[2]

Irregular movement

Real-life scenarios

Situations change frequently

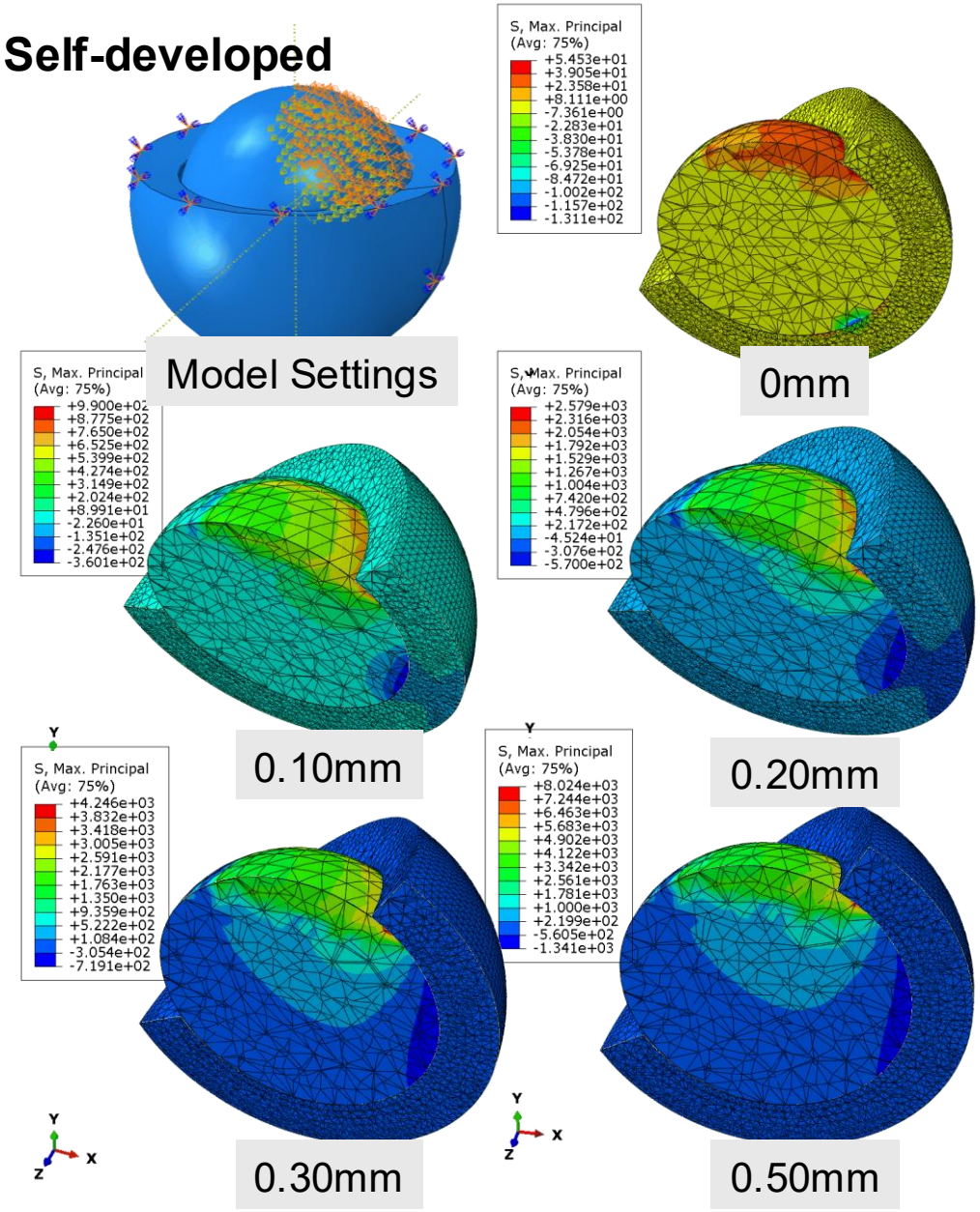
- Micro-separation
- Edge contact
- Impact loads
- Joint misalignment or installation angle deviation
- Surface scratches and polishing damage
- Abrasion particles
- Wet-heat aging
- Cyclic fatigue
- Variations in patient gait

Formation of localized tensile stress hotspots

Xuanming Che	
Key Issues	

Interactive FEA: microseparation → stress concentration

Self-developed



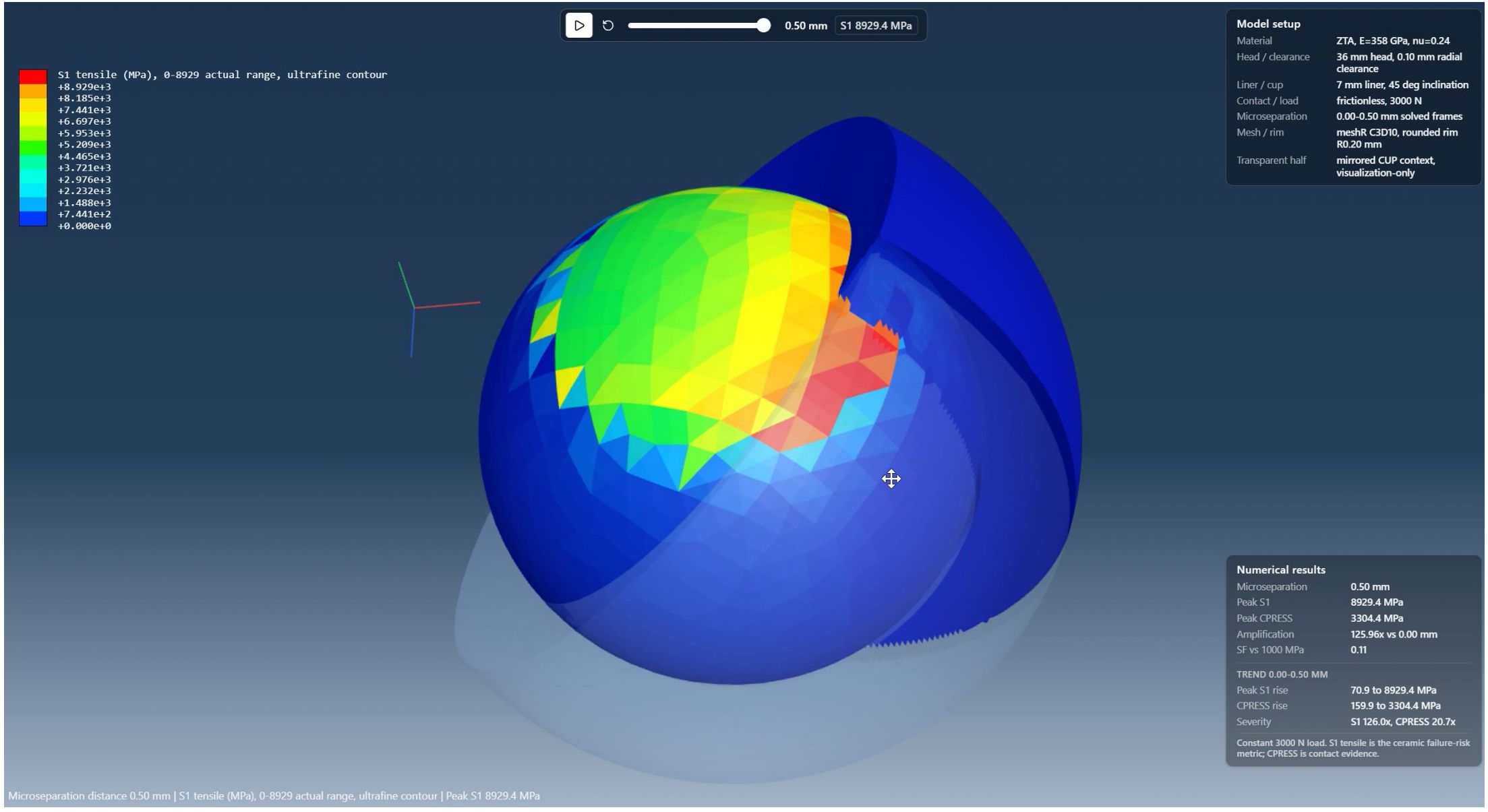
Model Setup	
Material	ZTA, E = 358 GPa, nu = 0.24
Geometry	36 mm femoral head, 0.10 mm radial clearance
Cup	7 mm liner, 45 deg inclination, rounded rim R0.20 mm
Contact/load	frictionless surface-to-surface contact, constant 3000 N
Parameter	microseparation distance: 0.00 to 0.50 mm

Microseparation Distance（mm）	S1 Tensile（MPa）
0	70.9
0.10	1178
0.20	3041
0.30	4970
0.50	8929

Boundary singularity? (Too big)

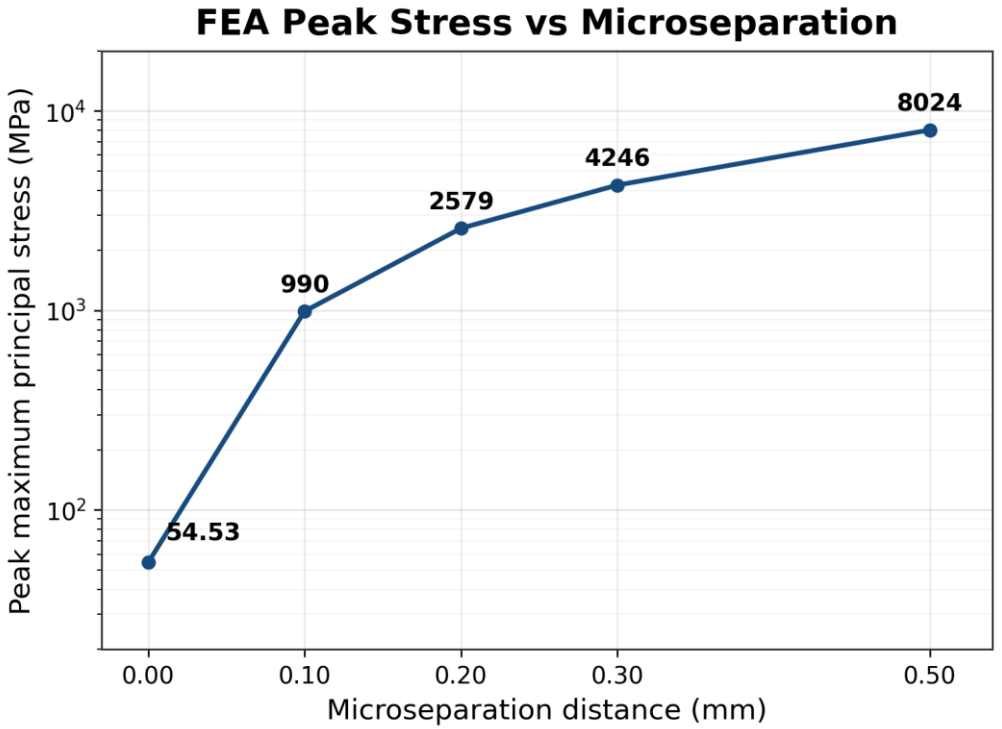
Xuanming Che	
Key Issues	

Interactive FEA: microseparation → stress concentration



Xuanming Che	
Key Issues	

Can local stress hotspots, microstructural defects and Weibull statistics be coupled into a patient-specific failure-risk map?



Increase in micro-separation distance

↓

Rapid rise in maximum principal tensile stress

↓

Formation of localized stress hotspots

The FEA provides the stress field

$\sigma_{1,\max}$ rises sharply as the degree of micro-separation increases.

Microstructural features result in uneven strength distribution

Key factors: pores, agglomerates, polishing cracks, residual stress, grain size 。

The Weibull distribution converts stress into risk

Failure-risk indicator $P_f = 1 - \exp\left[-\left(\frac{\sigma}{\sigma_0}\right)^m\right]$

The higher the σ_1 value, the larger the stressed volume, or the lower the m-value → the higher the probability of failure.

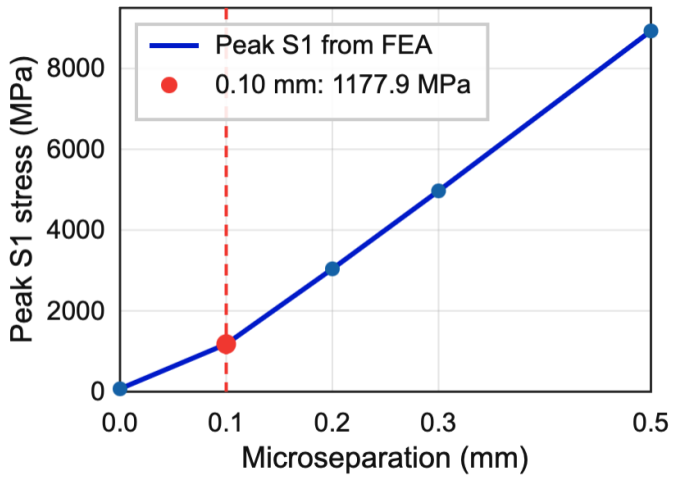
Xuanming Che	
Key Issues	

Interactive: Can local stress hotspots, microstructural defects and Weibull statistics be coupled into a patient-specific failure-risk map?

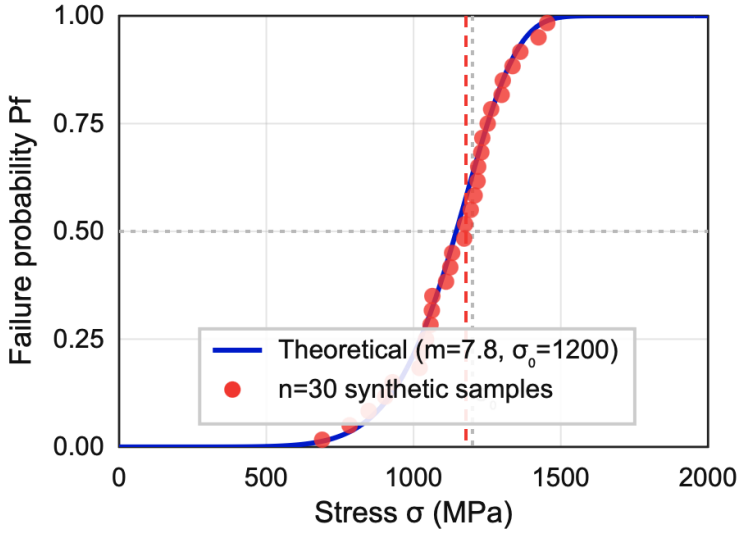
TYPICAL m: sintered structural ceramic

Tang et al. 2012: ZTA20 m = 7.8

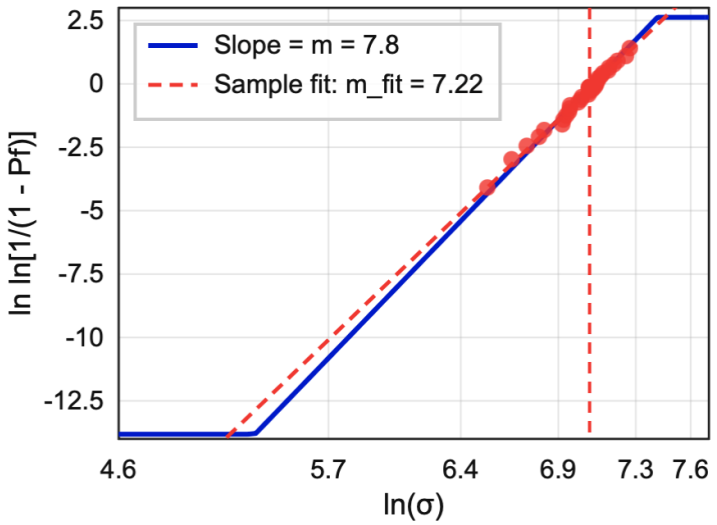
FEA stress hotspot



Weibull CDF + FEA stress



Weibull plot — slope = m



microsep.	0.10 mm	1177.9 MPa
m		7.8
σ_0 (MPa)		1200
n samples		30

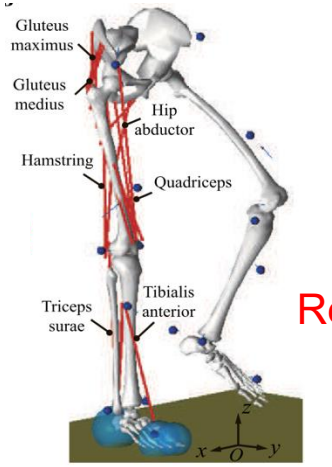
Not just materials, but a whole system:

Personalized Service + End-to-End Tracking

EMG sensors collect information

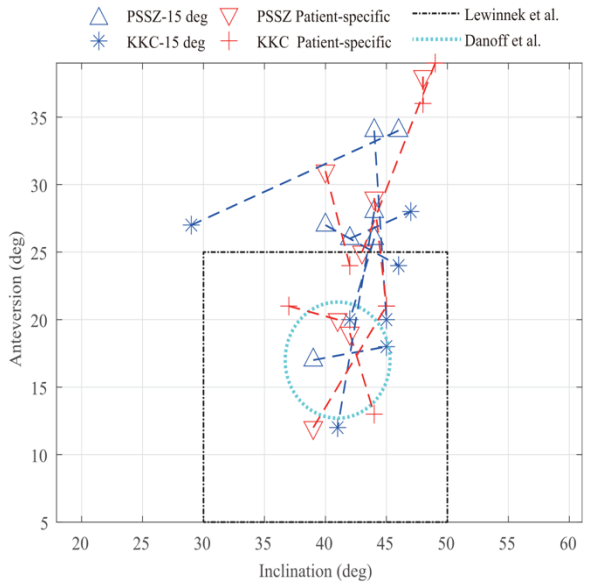


Musculoskeletal Model



Reaction force

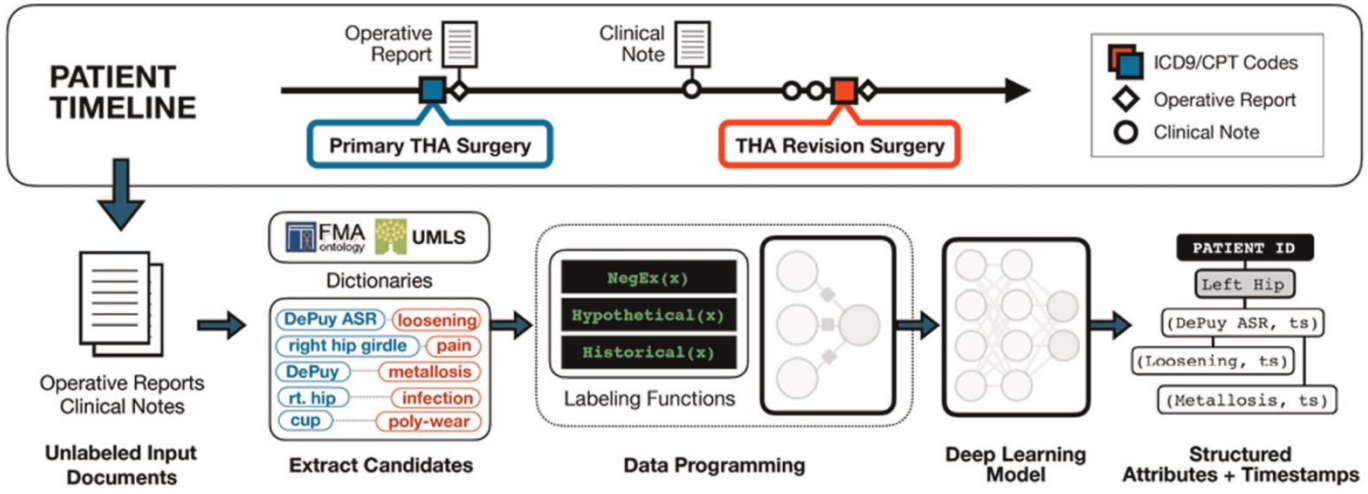


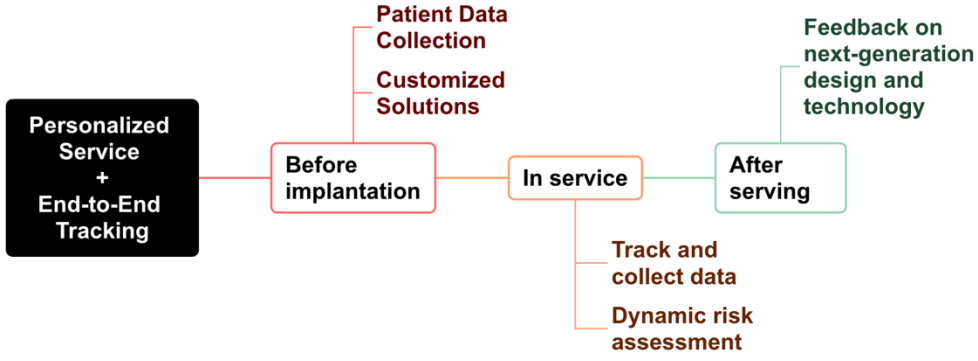


∗+ indicates the final recommended result optimized for biomechanics; the dashed line indicates the traditional safety zone. Individualized acetabular cup positioning results vary for each patient.[5]

Optimizing acetabular cup positioning for each patient

Personalised service: Gait analysis using EMG sensors to determine the range of motion through musculoskeletal simulation[5]

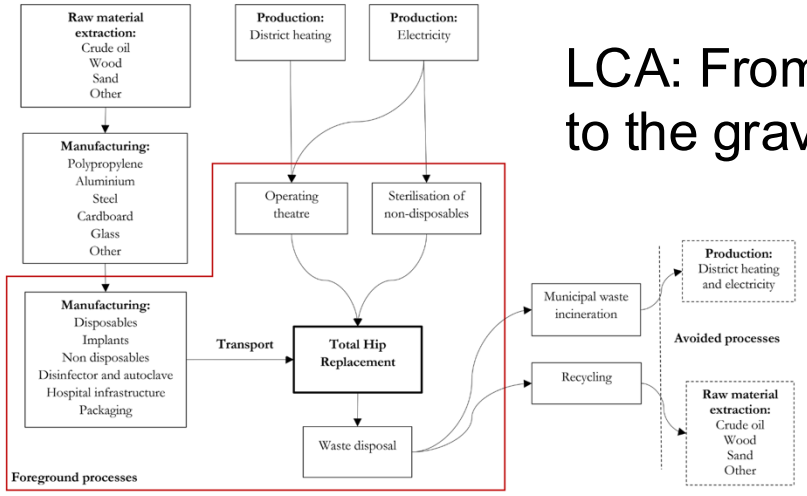




End-to-end Tracking: Record information on hip prosthesis models, complications, and pain, and use deep learning models to evaluate clinical outcomes and conduct digital health tracking.[6]

Personalized Care and End-to-End Monitoring

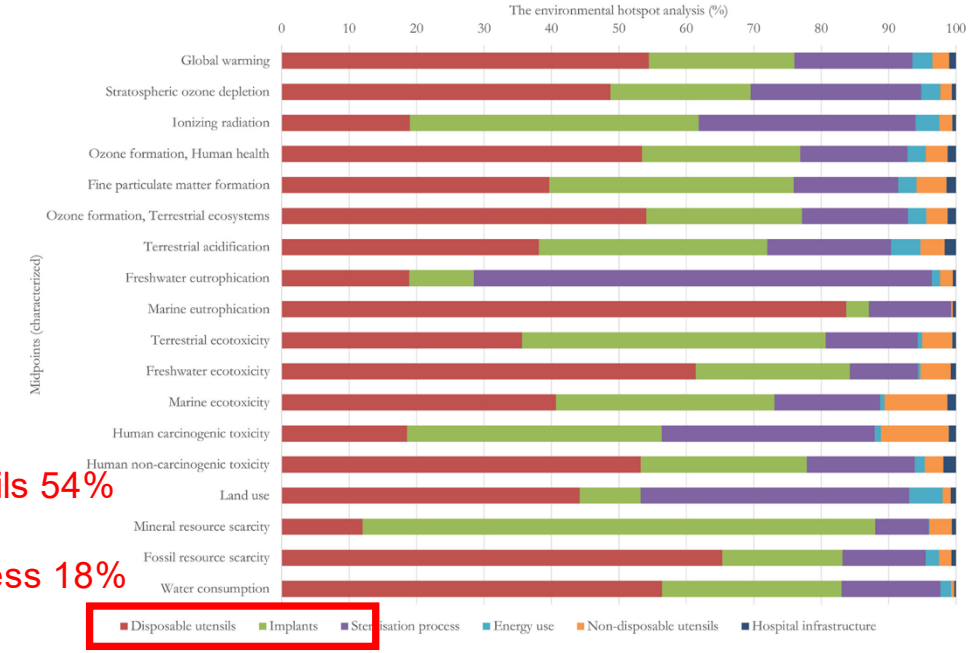
Sustainability and Green Design: Extending the Lifespan of the ZTA Hip Prosthesis and Reducing Its Environmental Impact



System Boundaries for the Life Cycle Assessment of Total Hip Replacement[7]

LCA: From the cradle to the grave

- Disposable utensils 54%
- Implants 22%
- Sterilisation process 18%



The environmental impact of a total hip replacement normalised to a scale of 100% (Midpoint Scores).[7]

The total carbon footprint of THR is 62.0 kg CO_2e per procedure.

Developing Risk-Adjusted Sustainability Indicators

$$Environmental\ burden\ per\ patient\text{–}year = \frac{E_{manufacturing} + E_{QA} + P_{revision} \times E_{revision}}{reliable\ service\ life}$$

Short-term: Optimize sterilization and surgical procedures
Medium-term: Establish a ZTA quality passport
Long-term: Patient-specific digital twins + lifespan prediction

Reference by Haotai Xu

1.Ning J., Ren H., editors. Mechanical Properties and Constitutive Model of Toughening Ceramics2015 2015.

2.Chen M. W., McCauley J. W., Dandekar D. P., Bourne N. K. Dynamic plasticity and failure of high-purity alumina under shock loading. Nature Materials. 2006;5(8):614-8.

3.Xu Zejian, Cai Zhicheng, Han Yang, Igusheva Liudmila, Petrov Yuri, Zhao Shixiang, et al. Fracture toughness and failure mechanism of alumina and zirconia ceramics over a wide loading rate range. International Journal of Impact Engineering. 2026;209:105591.

4.Yin Shuang, Sun Chengyun, Li Xiang, Zhang Yao, Fang Xia, Ma Haiqiang, et al. Phase composition, microstructure, and mechanical properties of ZTA ceramics prepared by gelcasting using multi-gradation spherical powders. Journal of Alloys and Compounds. 2025;1017:178984.

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6.Zhu W. Z. Grain size dependence of the transformation temperature of tetragonal to monoclinic phase in ZrO2(Y2O3) ceramics. Ceramics International. 1996;22(5):389-95.

Reference by Xiangjun Zheng

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[2] CESARI F, ESPOSITO L, FURGIUELE F M, et al. Fracture toughness of alumina–zirconia composites[J]. Ceramics International, 2006, 32(3): 249-255.
[3] KOLTSOV I, KIMMEL G, STELMAKH S, et al. The new nano-enabled phase map of ZrO_2-Al_2O_3[J]. Scientific Reports, 2019, 9: 5540. https://doi.org/10.1038/s41598-019-42058-4
[4] SHENG S H, ZHANG R F, VEPREK S. Study of spinodal decomposition and formation of nc-Al_2O_3/ZrO_2 nanocomposites by combined ab initio density functional theory and thermodynamic modeling[J]. Acta Materialia, 2011, 59(9): 3498-3509.
[5] RITTIDECH A, SUEKWAMSUE N. Influence of yttria additives on structural, microstructural and mechanical properties of alumina–zirconia composites prepared by two-stage sintering[J]. Ceramics International, 2015, 41(S1): S123-S126.
[6] FABRICHNAYA O, SAVINYKH G, SCHREIBER G. Phase relations in the ZrO_2-La_2O_3-Y_2O_3-Al_2O_3 system[J]. Journal of the European Ceramic Society, 2013, 33(13): 2689-2702.

Reference by Yongqi Xu

1. Zhang B, Zhao H, Wang X, Li B, Yan X, Lin X. Effect of organic binders on physicochemical characteristics and sintering perfo rmance for spray-dried ZTA ceramic powder. J Alloys Compd. 2026 May 11. doi: 10.1016/j.jallcom.2026.179635.
2. Naik AK, Siddharth, Sahoo SN, Roy M, Roy S. Development of functionally graded alumina–zirconia composites by pressureless sintering for biomedical implant applications. Ceram Int. 2026;52(9 Part B):13242-13253.
3. Liu J, Yin S, Sun C, Chen C, Zhang Y, Liu L, et al. Multi-scale/multi-morphology composite powders synergistic packing for solid loading regulation of gelcasted ZTA ceramics. Ceram Int. 2026 Mar 4. doi: 10.1016/j.ceramint.2026.03.032.
4. Zhang B, Lin X, Yan X, Luo J, Li Z, Cheng B, et al. Microwave-sintered graphene/ZTA biocomposites for artificial joints: Synergistic mechanical reinforcement, tribological enhancement, and biofunctional improvement. J Mater Res Technol. 2026 Apr 19. doi: 10.1016/j.jmrt.2026.03.121.
5. Maletskii AV, Isayev RS, Volkova GK, Belichko DR, Zaporozhets VD. Microstructure-governed response of ZTA ceramics to proton irradiation: From degradation to radiation-stimulated reorganization. Ceram Int. 2025;51(27 Part A):52271-52283.
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Thanks

This was the last major coursework assignment I completed at the Chang'an Campus. This course was unique because it didn’t involve traditional exams; instead, it allowed us to truly immerse ourselves in materials science problems through major assignments, which gave me a deeper appreciation for the inherent appeal of scientific research. Although my future research may lean more toward polymers and may no longer be directly related to ceramics (lol), the training and inspiration I gained from this course remain invaluable.

I am deeply grateful to Professors Grasso and Sun for their meticulous design of this course. It was one of my favorite courses during my time at QM. We benefited immensely from Professor Grasso’s rigorous and meticulous scientific approach, as well as the many insights Professor Sun shared this semester regarding current research hotspots and cutting-edge developments.

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P.S. The narrative logic of this report is quite different, and we encountered many difficulties during the production process, but I truly appreciate your words of encouragement, which gave us the motivation to keep going.

Please keep up your enthusiasm and follow the option B as it sounds more exciting.

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